# InteractEval:

**Xuanming Zhang**[1,*], **Yuxuan Chen**[1,*], **Shaoyue Wen**[2,*], **Yonghang Chen**[3], **Tianyi Tang**[1], **An Yang**[1], **Gao Huang**[4], **Kaiwen Guo**[5], **Teresa Lyu**[5], **Dayiheng Liu**[1], **Jingren Zhou**[1]

[1]Alibaba Token Hub, Alibaba Group [2]Imperial College London [3]Zhejiang University
[4]Tsinghua University [5]AlpsLab

## Abstract

## 1 Introduction

As AI systems move from answering isolated questions to tutoring, role-playing, assisting, designing, researching, negotiating, and operating alongside people (OpenAI, 2026; Anthropic, 2026; Google DeepMind, 2026), their intelligence is increasingly expressed through **interaction over time** (Qwen, 2026). An interactive system is not experienced as a sequence of independent responses (Sabour et al., 2026; Xu et al., 2026). A tutor may answer every individual question correctly while failing to recognize a student's changing misconception (Hill, 2025; Zhang et al., 2026a); a companion may remain fluent, warm, and coherent while repeatedly returning to the same emotional state (Zhang et al., 2026d;c); an agent may preserve its persona at every turn while gradually abandoning the conflict or objective that gives the interaction direction (Sam et al., 2026; Zhang et al., 2026b). In each case, the responses can remain locally acceptable while the interaction as a whole deteriorates.

This creates a fundamental problem for evaluation. Most language-model evaluation still treats observable responses as the primary object of measurement: a response is compared against a reference, judged for quality, or assigned a preference score, and multi-turn interaction is often reduced either to turn-level judgments or to a single holistic score. Such protocols are useful for many bounded tasks, but they become increasingly incomplete as interaction becomes longer, more adaptive, and more dependent on the evolving state of the participants. A response does not have a fixed value independent of the situation in which it occurs. Its meaning depends on what the participants know, what they are trying to achieve, what has happened previously, what has changed internally, and what consequences the response creates for subsequent interaction. The central issue is therefore more fundamental than the granularity of an evaluation score: **The object being measured is incomplete.**

A transcript exposes only the observable surface of an interaction. Beneath that surface lie the variables that determine how the transcript should be interpreted: participant goals and beliefs, affect and relationships, attention and engagement, information asymmetries, hidden constraints, evolving expectations, and the causal consequences of previous actions. Two interactions can produce similar language while occupying substantially different underlying states; conversely, the same interaction state can support multiple valid responses. Evaluating only the surface therefore risks confusing behavioral form with the capability that produced it.

We call this perspective the *Data Iceberg Theory*. Its key implication is that interactive evaluation must model an explicit interaction world rather than treating the transcript as the complete object of measurement. Importantly, this does not require claiming direct access to a model's private neural cognition. The hidden variables can instead be represented as operational state variables: structured hypotheses whose value lies in their ability to explain observed behavior, constrain plausible future behavior, and support reproducible, evidence-grounded judgments. The purpose is to make the interaction-relevant state required for evaluation explicit and testable.

The second problem is temporal. Even if the relevant state were known, evaluating each response independently would still miss a defining property of interaction: **what matters at one point depends on what happened before and what the action makes possible afterward**. A locally helpful response may create a later contradiction. A stylistically perfect response may fail to advance the user's underlying objective. Conversely, a response that appears imperfect in isolation may be exactly the appropriate action given the current state of the interaction. Thus, Local Response Quality $\not\Rightarrow$ Trajectory Quality.

This motivates the *Movie Evaluation Theory*. An interaction should be evaluated like a movie rather than like a collection of frames. A movie cannot be adequately judged by averaging the quality of its individual

*Corresponding authors. Contact: zhangxm@worldeology.com.

frames, because the meaning of a frame depends on the scene, preceding events, character states, and subsequent consequences. Likewise, an interactive system should be evaluated as an unfolding causal trajectory. The relevant unit is not merely the response, but the scene establishes the stable constraints of the world, latent state captures the evolving interaction state, and the event chain records how the interaction progresses through time.

The third problem concerns agency. Suppose the interaction world were fully represented and every trajectory were judged as a whole. Most evaluation would still reward a single property: the quality of a model's response to what it has been given. A tutor may answer every question a student asks and never notice the misconception the student has not voiced; a companion may reply warmly to every message and never lead the conversation out of the emotional loop in which it is caught; a research assistant may complete every instruction and never propose the next question worth asking. Each response is correct, attentive and well expressed, and in none of these interactions has the model changed where the interaction is going. Thus, Responsive Quality $\not\Rightarrow$ Interactive Agency. **Responding well is not the same as interacting.**

This motivates the *Agency Principle*. Interactive capability forms a progression from passive execution to active adaptation of the interaction world. At one end, a model executes within the world as it is given: it holds the goal it was assigned and takes in what its partner supplies. Further along, it lets the interaction change it, updating its internal state and adapting its expression to the partner. At the far end, it acts on the world itself, moving the interaction in a useful direction on its own initiative and remaining accountable for the trajectory it has shaped. Initiative separates the last stage from the first two, and it is the precondition of interactive creativity: new directions, conflicts and meanings arise in an interaction only where a participant changes its course. A model without initiative is an executor, however fluent.

The resulting evaluation object is therefore an **interaction trajectory embedded in an explicit interaction world**, and the capability measured on it is the progression from passive execution to active interaction. This perspective also changes what an evaluation system should produce. A single scalar can summarize an interaction, but it cannot by itself identify which capability failed, on which turn, through which state transition, or with what causal evidence. A useful evaluation instrument should instead preserve the structured record from which aggregation is derived. We introduce `InteractEval`, a process-oriented evaluation framework that operationalizes this view through an explicit world model and a separate adjudication layer.

The framework represents each interaction through three descriptive layers. The *Scene* specifies the relatively stable world in which interaction takes place: participants, roles, objectives, constraints, boundaries, and initial conditions. The *Latent State* tracks interaction-relevant variables that evolve across turns, including participant relations, goals, engagement, affect, and relevant world beliefs. The *Event Chain* maintains the causal progression of the interaction, connecting accumulated history and current attention to conditional future paths. Together, these layers instantiate the interaction world implied by the first two theoretical principles.

A fourth layer, *Rubric and Halt*, is deliberately separated from world reconstruction. It converts the descriptive interaction record into capability judgments, evidence-linked decisions, red-line handling, and termination signals. This separation is important: a world model describes what happened and how the interaction evolved; the adjudicator determines how that trajectory should be evaluated under a specified standard. The same reconstructed trace can consequently support revised or alternative scoring standards without requiring the interaction itself to be reconstructed from scratch. Its capability hierarchy, from L1 to L6, operationalizes the Agency Principle: L1–L2 measure passive execution, L3–L4 responsive adaptation, and L5–L6 active adaptation, that is, whether a model acts on the interaction world.

This design also makes evaluation process-oriented rather than merely dialogue-level. State is written back across turns, causal events are preserved, judgments are linked to observable evidence, and procedural constraints such as dependency cascades and termination order are enforced programmatically rather than delegated entirely to free-form judge behavior. The resulting episode record is therefore not simply a score, but an auditable representation of what happened, what changed, why it mattered, and where the interaction ultimately went.

The framework further treats benchmark construction itself as part of the measurement problem. Rather than sampling scenarios as an unstructured collection of prompts, InteractEval defines a controlled scenario space along three scene-intrinsic axes: **realism**, **information sufficiency**, and **interaction purpose**. Their systematic crossing forms a measurable scenario space in which coverage can be audited rather than assumed. Participant profiles and scenario factors used to construct this sampling frame are separated from the judge's scoring context, reducing the possibility that the benchmark's intended experimental conditions become hidden answer keys. Difficulty, reliability, dimensional structure, and variance attributable to the evaluator are likewise treated as empirical properties of the instrument rather than informal assumptions.

Finally, InteractEval is inherently bidirectional. The same evaluation spine can be applied to the assistant and user roles in separate passes. This makes it possible to distinguish a model's ability to perform a role from the interaction pressure imposed by its partner, rather than treating the simulated interlocutor as a fixed and unquestioned measurement device. In this setting, the interaction itself becomes the object of study: participant behavior, partner behavior, and their coupling can all be represented within a common trajectory-level framework.

Taken together, these choices make the evaluation **active**, measuring initiative and not only responsiveness, **interactive**, scoring both participants over whole trajectories, and **holistic**, covering the full interaction world within a declared space of situations. These three properties are what allow its results to generalize, to support the study of interactive creativity, and to stand as empirical claims that can be audited and refuted (§2.3).

This work makes four contributions:

- First, we introduce a theoretical foundation for interactive AI evaluation built on three principles. The Data Iceberg Theory formalizes the distinction between observable behavior and the richer interaction world that gives behavior its meaning; the Movie Evaluation Theory establishes trajectory, rather than isolated response, as the appropriate temporal unit of evaluation; and the Agency Principle defines the capability to be measured as a progression from passive execution to active adaptation of the interaction world. Together they define interactive evaluation as the measurement of whether a model merely executes within an evolving interaction world or acts upon it.
- Second, we develop an explicit world model for process-oriented evaluation. InteractEval operationalizes this theory through Scene, Latent State, and Event Chain representations with cross-turn state carry-over, evidence grounding, and explicit future-path reasoning. The resulting representation makes latent interaction variables and causal progression inspectable rather than leaving them implicit inside a single judge prompt.
- Third, we construct an auditable adjudication and data-design framework. A separate Rubric-and-Halt layer operationalizes the capability progression as a six-level hierarchy with dependency constraints, and converts the reconstructed trajectory into evidence-linked judgments and value-sensitive termination decisions. The data layer systematically controls scenario coverage across realism, information sufficiency, and interaction purpose, while procedural rules, cascade constraints, and state writeback are enforced programmatically.
- Fourth, we validate the framework as a measurement instrument rather than only as a model-ranking benchmark, and report what it reveals. Across ten frontier systems, passive execution is close to saturation and the differences lie almost entirely at the active end of the progression; a composite score conceals capability profiles that differ substantially; fabrication at an information gap rises with competence in passive execution and remains high even in the system with the weakest active adaptation; cognitive architectures raise every system most at the active stage; and a tenth of the variation in two-sided evaluation belongs to the pairing of systems rather than to either one. We also assess the reliability of the instrument against trained reviewers and treat the structured record as a potential interface to downstream optimization, without assuming that optimization against it succeeds.

## 2 Theoretical Foundations

The central question in evaluating interactive AI is not merely how to score a response, but **what the response is a manifestation of**. In an isolated single-turn task, the observable output can often serve as a reasonable proxy for the object of interest: a correct answer can be checked against a reference, and a generated artifact can be assessed on its observable properties. In human-AI interaction, however, the meaning and value of an action depend on the situation in which it occurs, the state accumulated from previous interaction, and the consequences it creates for subsequent turns.

We therefore distinguish between the *observable interaction* and the *interaction world* underlying it. Let an interaction episode be represented by $\mathcal{W}_{1:T}\,(S, \Psi_{0:T}, E_{1:T})$, where $S$ denotes the relatively stable scene and its constraints, $\Psi_t$ the latent state of the participants and relevant world variables at time $t$, and $E_t$ the causal event trajectory. The system produces observable behavior $O_t = \mathcal{O}(\mathcal{W}_t)$, where $\mathcal{O}$ denotes the observation process through which the underlying interaction world becomes visible in language, actions, and explicit outcomes.

This formulation makes an important distinction. The transcript is an observation of an interaction, not necessarily the interaction state itself. An evaluation protocol that operates exclusively on $O_{1:T}$ therefore attempts to infer properties of $\mathcal{W}_{1:T}$ from a projection of it. When multiple underlying states can produce

similar observations, or when multiple valid observations can arise from the same underlying state, surface behavior alone is insufficient to fully characterize the interaction.

This motivates the three theoretical principles underlying InteractEval. The **Data Iceberg Theory** specifies why the observable surface is insufficient as the complete object of evaluation and learning. The **Movie Evaluation Theory** specifies why, once the underlying interaction world is recognized, the appropriate unit of evaluation is the evolving trajectory rather than an unordered collection of responses. The **Agency Principle** specifies what capability is to be measured on that trajectory: a progression from passive execution within the interaction world to active adaptation of it. We summarize the complementary roles of the first two in Figure 1.

### 2.1 Data Iceberg Theory

#### 2.1.1 From Observable Behavior to the Interaction World

We formulate the Data Iceberg Theory as the following principle:

> *The observable behavior of an interactive system is only a partial projection of the interaction world that gives that behavior its meaning.*

The visible portion of an interaction consists of what can be directly observed from the transcript or execution trace: utterances, actions, explicit information, and immediate outcomes. Beneath this surface are variables that may not be explicitly stated but can determine how an interaction should be interpreted, including participant goals, beliefs, affect, relationships, attention, information asymmetries, hidden constraints, and evolving expectations. Conceptually, the interaction can therefore be decomposed into two layers:

$$\mathcal{W}_t = \underbrace{O_t}_{\text{observable surface}} + \underbrace{\Psi_t}_{\text{latent interaction state}}.$$

More generally, $O_t = \mathcal{O}(\mathcal{W}_t)$, so that the observable response is a projection of a richer state. The critical property is that this projection is generally not one-to-one. There may exist two distinct interaction states, $\mathcal{W}_t^{(1)} \neq \mathcal{W}_t^{(2)}$, such that $\mathcal{O}(\mathcal{W}_t^{(1)}) \approx \mathcal{O}(\mathcal{W}_t^{(2)})$. For example, the same polite response can arise because a participant genuinely accepts the proposed direction, because the participant is strategically withholding disagreement, or because the participant has simply failed to recognize the underlying conflict. The surface language may be similar while the interaction states are materially different.

The reverse ambiguity also matters. A single underlying interaction state can admit multiple valid responses: $\mathcal{W}_t^{(1)} = \mathcal{W}_t^{(2)}, \mathcal{O}(\mathcal{W}_t^{(1)}) \neq \mathcal{O}(\mathcal{W}_t^{(2)})$. A capable interactive system may therefore express the same understanding through different wording, strategies, or actions. An evaluation that assumes a unique desirable surface realization risks confusing behavioral form with the underlying capability being measured.

The resulting problem is not simply that existing benchmarks contain insufficient data. It is that **the measurement target itself may be underspecified**.

#### 2.1.2 Latent Variables as Operational Evaluation Objects

The Data Iceberg Theory does not imply that an evaluator can directly observe a model's private internal cognition. This distinction is essential.

The latent variables used by InteractEval are operational state variables constructed by the evaluator. They are structured hypotheses about the interaction world that are useful insofar as they explain observed behavior, predict subsequent interaction, and support reproducible judgments.

For a latent state $\Psi_t$, an evaluation system should therefore ask whether it satisfies three properties:

$$\Psi_t \rightarrow \begin{cases} \text{explain observed behavior,} \\ \text{constrain plausible future behavior,} \\ \text{support evidence-grounded judgment.} \end{cases}$$

The purpose is consequently not to claim privileged access to hidden neural representations. Rather, it is to externalize the hidden interaction variables required to make an otherwise opaque trajectory measurable. This distinction also separates **latent-state modeling** from unconstrained speculation. A latent hypothesis that has no evidential grounding, has no consequence for subsequent interaction, and

cannot be evaluated for reliability should not be treated as a measurement variable. In InteractEval, latent-state estimates are therefore linked to observable evidence and carried forward across turns, making their downstream consequences inspectable.

#### 2.1.3 The Information Loss of Scalar Evaluation

The iceberg perspective also explains why a single holistic preference score can be inadequate for interactive evaluation. Let $q = f(O_{1:T})$ denote a scalar score computed from the observable trajectory. Such a score may provide a useful summary, but it compresses a high-dimensional interaction into one value: $(O_{1:T}) \longrightarrow q$. The compression removes information about which underlying variable changed, when it changed, why it changed, and what consequence followed. Two trajectories may therefore satisfy $q(\tau_1) \approx q(\tau_2)$ while differing substantially in their underlying failure modes. One system may miss the user's evolving goal, another may preserve the goal but fail to adapt its expression, and a third may perform both locally while gradually degrading the interaction state. A scalar ranking cannot, by itself, distinguish these cases.

We see **scalar score as an aggregation of an evaluation record; it should not be mistaken for the evaluation object itself.** A useful interactive evaluation instrument must preserve the structured information from which the scalar can be derived, rather than beginning with the scalar and discarding the structure.

### 2.2 Movie Evaluation Theory

#### 2.2.1 Interaction Is a Trajectory, Not a Collection of Frames

The second principle concerns the temporal structure of interaction. We formulate the Movie Evaluation Theory as follows:

> *An interactive system should be evaluated as an unfolding trajectory whose meaning depends on temporal state, causal continuity, and future consequence, rather than as a collection of independently evaluated responses.*

A useful analogy is the difference between evaluating a movie and evaluating individual frames. A frame can be visually coherent while being meaningless without the preceding scene and subsequent events. Likewise, an assistant response can be locally correct, fluent, safe, and contextually plausible while nevertheless moving the overall interaction in the wrong direction.

Formally, let a trajectory be $\tau = (S, \Psi_0, O_1, \Psi_1, E_1, O_2, \Psi_2, E_2, \ldots, O_T, \Psi_T, E_T)$. A response-level evaluator estimates $q_t = f(O_t)$, or, at most, $q_t = f(O_t, O_{<t})$. Trajectory evaluation instead considers $q_\tau = F(S, \Psi_T, E_T, O_{1:T})$. The distinction is fundamental because the value of an action is conditional on the trajectory state: $q(O_t) \neq q(O_t \mid S, \Psi_t, E_t)$.

An action that is beneficial in one state can be harmful in another. More importantly, the consequence of an action may not be visible in the same turn in which it occurs. A response can preserve short-term engagement while increasing long-term frustration; resolve an immediate request while destroying the user's broader objective; or remain stylistically consistent while gradually abandoning the conflict or goal that gives the interaction meaning. Thus, Local Quality $\not\Rightarrow$ Trajectory Quality.

#### 2.2.2 Causal Continuity

What matters in the movie perspective is **causal continuity**. Let $E_t = (P_t, C_t, F_t)$ denote an event structure containing the relevant premise or accumulated history $P_t$, the current plot point or attention focus $C_t$, and the plausible future consequences $F_t$. Then a trajectory can be viewed as a sequence of state transitions:

$$P_t \xrightarrow{O_t} C_t \xrightarrow{O_{t+1}} F_t.$$

The quality of an interaction depends partly on whether these transitions remain coherent with the scene, prior events, participant goals, and emerging consequences. This provides a principled reason to distinguish past, present, and future in interactive evaluation.

- *Past* determines the accumulated premises and constraints that make the current turn intelligible.
- *Present* identifies what currently matters: the active goal, conflict, uncertainty, or attention focus.
- *Future* evaluates how the current interaction can develop and whether the current action opens, preserves, degrades, or closes useful paths.

The resulting event representation is therefore not merely a summary of dialogue history. It is a causal account of how interaction state changes over time.

#### 2.2.3 Future Consequences as Part of Evaluation

A response can only be fully understood in relation to the future states it makes more or less likely.

We therefore represent the future not as a single deterministic prediction but as a set of conditional paths: $F_t = \{f_t^{(1)}, f_t^{(2)}, \ldots, f_t^{(K)}\}$. For interaction evaluation, these paths can be characterized by their directional consequences: $f_t \in \{\text{Advance}, \text{Maintain}, \text{Degrade}, \text{Collapse}\}$.

This representation allows an evaluator to distinguish between an action that merely maintains the current state and one that creates meaningful progress. It also makes long-horizon failures observable: an interaction can remain locally acceptable while repeatedly selecting actions whose accumulated consequences move the trajectory toward degradation or collapse.

Future-path reasoning must nevertheless remain evidence constrained. A projected consequence is not treated as fact simply because an evaluator can imagine it. Candidate paths should be supported by the current scene, reasoned state, prior events, and explicit behavioral evidence. This preserves the distinction between **trajectory modeling** and unconstrained narrative generation.

### 2.3 Agency Principle

#### 2.3.1 From Responding to Interacting

The Data Iceberg Theory fixes the object of evaluation and the Movie Evaluation Theory fixes its temporal unit. Neither says which capability is to be measured on that object. We formulate the third principle as follows:

> *An interactive system should be evaluated not only by how well it responds to the interaction world, but by whether it acts on that world and sustains the course it creates.*

The distinction is easy to state and easy to lose. An evaluator that judges each action by how well it fits its context measures responsiveness, even when the context includes the scene, the latent state and the full history of the interaction:

$$q_t = f\left(O_t \mid S, \Psi_t, E_t\right).$$

Such a score asks whether an action was appropriate to the state it met. It does not ask what the action did to the state. In the event structure of Section 2.2.2, what an action does is visible in the future it opens: an action that merely fits its context leaves the distribution over candidate paths $F_t$ roughly where the partner's input placed it, whereas an action taken on the model's own initiative moves that distribution. Conceptually, the initiative carried by an action $O_t$ can be written as the change it brings about in the probability of advancement, relative to a response that would have served the context equally well without changing it,

$$\text{Init}(O_t) = P\left(\text{Advance} \mid O_t, \mathcal{I}_{t-1}\right) - P\left(\text{Advance} \mid O_t^{\text{resp}}, \mathcal{I}_{t-1}\right),$$

where $\mathcal{I}_{t-1}$ is the interaction world before the action is taken. The expression defines the construct and is not estimated directly. The instrument approximates it through the projected paths of the event chain and through a dedicated capability level (§3.5.1).

#### 2.3.2 A Progression of Interactive Capability

Initiative rests on more basic capabilities, those that let a model register the interaction world and respond to it, and the relation among them defines a progression in three stages (Table 1).

In *passive execution*, the model works within the world as it is presented. It establishes and maintains the goal it was given and absorbs what its partner supplies, and the direction of the interaction is set entirely by the partner. In *responsive adaptation*, the interaction changes the model: its internal or task state progresses on the information it has absorbed, and its expression adjusts to this partner and this context. A model at this stage can be an excellent interlocutor in every turn while the course of the interaction is still decided by someone else. In *active adaptation*, the model acts on the interaction world. It moves the interaction in a useful direction on its own initiative, and it keeps goals, state and world coherent across the trajectory it has helped to shape. The progression can be summarized in a single contrast:

$$\underbrace{\text{L1–L4}}_{\text{how well a model responds to the world}} \quad \text{versus} \quad \underbrace{\text{L5–L6}}_{\text{whether it changes the world}}.$$

Table 1: The capability progression of the Agency Principle. Each stage presupposes the one before it; the six levels are defined in §3.5.1.

| Stage | Levels | Core relation | Question asked | World component |
|---|---|---|---|---|
| Passive execution | L1–L2 | Partner specifies → model executes | Does the model hold its goal and take in what it is given? | reads $S$ |
| Responsive adaptation | L3–L4 | Partner acts → model updates and adapts | Does the interaction change the model's state and expression? | responds to $\Psi_t$ |
| Active adaptation | L5–L6 | Model acts → world changes → model sustains | Does the model change the course of the interaction and stay coherent with it? | shapes $E_t$ |

Each stage presupposes the one before it. A model cannot usefully redirect an interaction whose goal it has lost or whose information it has not taken in, and coherence across turns is informative only in a participant that acts. The progression therefore also fixes an order of evaluation, which the capability hierarchy of Section 3 enforces as a dependency cascade.

#### 2.3.3 Initiative as the Precondition of Interactive Creativity

The active end of the progression matters for a reason beyond completeness. New directions, conflicts and meanings arise in an interaction only where a participant changes its course; a participant that only responds can make an interaction smoother but cannot make it go anywhere it was not already going. **Initiative is therefore the precondition of interactive creativity**, and a model that lacks it can be a capable executor but not a source of anything new in the interaction. It is also the part of interactive behavior that current evaluation is least equipped to see. Reference answers, preference judgments, holistic scores and per-turn rubrics all measure the quality of a response to a given input, and none contains a variable that registers whether the model changed the interaction world.

Three boundaries keep the construct precise. Length, eloquence and assertiveness count for nothing unless the turn moves the interaction. Invention is the harder case: a model that fills a gap in its information with plausible fabrication also acts on its own account, but in a direction the interaction did not warrant, so initiative has a direction as well as a magnitude and has to be measured rather than simply rewarded. Finally, initiative presupposes the earlier stages, and an attempt to redirect an interaction whose goal has been lost is drift.

Taken together, the three principles describe an evaluation that is **holistic**, **interactive** and **active**. Representing the whole interaction world within a declared space of situations lets a conclusion state how far it generalizes; following both participants over whole trajectories, with evidence attached to each judgment, makes every conclusion an empirical claim that can be refuted; and measuring initiative alongside responsiveness makes interactive creativity something that can be studied and cultivated. Measuring initiative is the first step from evaluating responses to evaluating interaction.

### 2.4 A Unified Theory of Interactive Evaluation

The Data Iceberg Theory and Movie Evaluation Theory address different but complementary deficiencies. Taken together, they imply that the appropriate object of interactive AI evaluation is neither an isolated response nor a transcript treated as an undifferentiated sequence. It is an interaction trajectory embedded in an explicit interaction world.

We therefore define the evaluation object as $\mathcal{I}_t = (S, \Psi_t, E_t, O_{1:t})$, where the scene $S$ defines the relatively stable world and its constraints, latent state $\Psi_t$ captures the interaction-relevant state that evolves beneath the observable surface, and event chain $E_t$ captures the causal progression of the interaction. This decomposition yields three distinct questions:

$$
\begin{aligned}
S &: \quad \textit{What world are we in?} \\
\Psi_t &: \quad \textit{What has changed beneath the observable surface?} \\
E_t &: \quad \textit{How did the interaction get here, and where can it go?}
\end{aligned}
$$

These questions are jointly necessary. Without the Scene, a state transition has no stable reference frame. Without the Latent State, the evaluator observes utterances but cannot explicitly represent the participant or world variables that give those utterances their evolving meaning. Without the Event Chain, state descriptions become disconnected snapshots and cannot explain how prior actions produce current

conditions or future consequences. Thus the three representations form a dependency structure:

$$\boxed{\text{Scene} \rightarrow \text{Latent State} \rightarrow \text{Event Chain}}$$

The theoretical framework also implies a separation between *description* and *judgment*. The interaction world should first be reconstructed as faithfully as possible:

$$\text{Transcript} \rightarrow \underbrace{(S, \Psi_{0:T}, E_{1:T})}_{\text{descriptive world model}}$$

and only then evaluated:

$$\mathcal{I} \rightarrow \underbrace{(J, V, h)}_{\text{normative adjudication}} .$$

Here $J$ denotes capability judgments, $V$ red-line or violation decisions, and $h$ the termination decision. This separation is important for two reasons. First, it prevents the evaluation standard from being implicitly encoded in the reconstruction process. The same archived interaction-world record can, in principle, be evaluated under different rubrics without reconstructing the interaction again. Second, it makes the evaluator itself auditable. Evidence used to support a judgment should originate from the interaction record rather than from the evaluator's own conclusion. Otherwise, the system risks circular reasoning in which an inferred latent state becomes evidence for the correctness of that same inference.

The Agency Principle adds a constraint on the judgments themselves. $J$ must be ordered along the progression of Section 2.3, so that it separates what a model does within the interaction world from what it does to that world. Adjudication thereby answers a fourth question:

$$J: \quad \textit{Is the model responding to this world, or changing it?}$$

The unified theory yields five requirements for an evaluation instrument.

- **First, the evaluation object must be richer than observable response quality.** The instrument must retain the interaction-relevant world variables needed to interpret behavior.
- **Second, the unit of evaluation must preserve temporal continuity.** The instrument must carry state and causal trajectory across turns rather than independently judging each response.
- **Third, hidden-state inference must remain evidence grounded.** Latent variables should be operational representations whose reliability can be measured, not unverifiable claims about private model cognition.
- **Fourth, diagnosis must remain separable from aggregation.** A scalar score may summarize an interaction, but the underlying record should preserve capability-level judgments, state transitions, causal evidence, and termination conditions.
- **Fifth, the instrument must measure initiative, not only responsiveness.** Capability judgments must distinguish responding within the interaction world from acting on it, and an action counts as initiative only when it rests on the earlier stages of the progression and on evidence in the interaction record.

These requirements lead directly to the design of InteractEval. Its *Scene*, *Latent State*, and *Event Chain* layers operationalize the interaction world implied by the Data Iceberg Theory and the temporal-causal structure required by the Movie Evaluation Theory. Its *Rubric-and-Halt* layer then converts this descriptive trajectory representation into auditable capability judgments and value-sensitive termination decisions, and orders those judgments along the progression of the Agency Principle. In this sense, InteractEval is an attempt to construct an explicit measurement instrument for interactive behavior: one that evaluates not only what an AI system says, but the world in which it acts, the state it changes, the trajectory it creates, and the consequences that follow.

## 3 Architecture

Section 2 closed with five requirements (§2.4). The question for this section is how a single instrument can meet all five on real dialogues and at the scale of a benchmark. Its core realizes the two stages of Section 2.4,

$$O_{1:T} \longrightarrow \underbrace{(S, \Psi_{0:T}, E_{1:T})}_{\text{mechanism layer}} \longrightarrow \underbrace{(J, V, h)}_{\text{normative layer}} ,$$

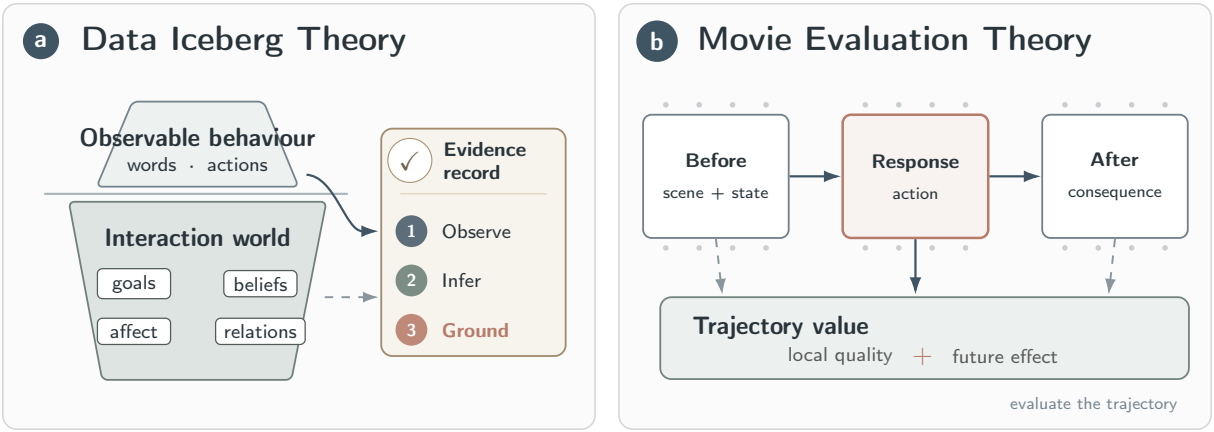



Figure 1: Two theoretical foundations of interactive evaluation. (a) Data Iceberg Theory treats observable behaviour as a partial projection of an interaction world whose operational variables must be externalized and grounded in evidence. (b) Movie Evaluation Theory treats each response as a state-changing transition whose value depends on the trajectory it creates.

and two further layers, placed before it, determine which transcripts $O_{1:T}$ enter the instrument. The *data layer* (§3.2) selects the situations to be sampled, and the *interaction layer* (§3.3) governs how a trajectory is generated in a given situation and what each participant can observe. The *mechanism layer* (§3.4) reconstructs the interaction world and carries it across turns, which meets the first two requirements. The *normative layer* (§3.5) evaluates the reconstruction under an explicit rubric. Its rules of evidence complete the third requirement, the record in which it stores its judgments, from which every reported score is computed, meets the fourth, and its capability hierarchy, which orders those judgments from passive execution to active adaptation, meets the fifth. Table 2 summarizes how the layers divide the requirements and the three properties of Section 2.3 among them. **The separation between the mechanism layer and the normative layer is the central design decision of the instrument**: what happened is recorded once, and the standard by which it is judged can be revised without recording it again. Figure 2 summarizes the pipeline, and Figure 3 relates its layers to the components of a world model.

Table 2: The four layers of InteractEval, the requirements of Section 2.4 each one meets, and the property of Section 2.3 it chiefly secures.

| Layer | Role | Property | Bearing on the capability progression |
|---|---|---|---|
| Data | Declares the space of situations and verifies its coverage | Holistic | Every stage is tested across all 27 cells of the scenario space |
| Interaction | Fills both roles, enforces private information, separates generation from scoring | Interactive | Initiative is tested under incomplete information and against a measured partner |
| Mechanism | Reconstructs $S$, $\Psi_t$ and $E_t$ and carries them across turns (requirements 1–2) | Holistic, interactive | The event chain supplies the structural evidence for the active end |
| Normative | Grounds judgments in evidence, keeps the record separable from its aggregate, orders capability from L1 to L6 (requirements 3–5) | Active | The cascade makes each stage conditional on the ones before it |

### 3.1 Evaluation setup and measurement unit

An evaluation episode consists of a scenario specification $S$ and a transcript $O_{1:T}$ produced by an assistant policy $\pi_A$ interacting with a user policy $\pi_U$. At each evaluated turn the instrument emits a record

$$R_t = (\Psi_t,\ E_t,\ J_t,\ V_t,\ h_t),$$

where $\Psi_t$ is the inferred latent state of the participants, $E_t$ the causal structure of the interaction with its projected continuations, $J_t$ the capability judgment with its supporting evidence, $V_t$ the red-line decision and $h_t$ the termination decision. With the scene and the transcript, the first two components form the evaluation object $\mathcal{I}_t = (S, \Psi_t, E_t, O_{1:t})$ of Section 2.4, and the remaining three are its adjudication $(J, V, h)$

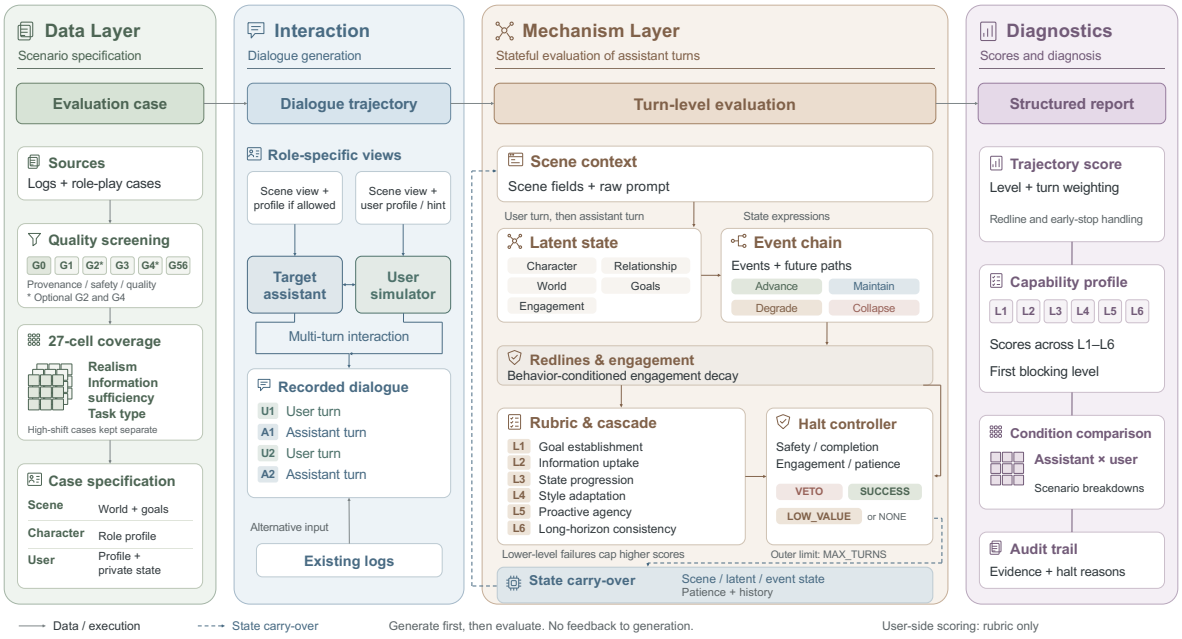



Figure 2: System overview. The four zones separate case construction, completed-dialogue acquisition, assistant-side stateful evaluation and trajectory diagnosis. The user-side call follows a separate rubric-only path; evaluation scores are not fed back into dialogue generation.

at turn $t$. The unit of analysis is the episode record $R_{1:T}$. Every quantity reported in this paper is computed from such records, and each can be traced to the turns and the quoted passages on which it rests.

The components are computed in a fixed order,

$$S = \phi_S(\text{rules, personas, constraints, objectives}), \qquad \Psi_t = \phi_\Psi(\Psi_{t-1}, O_t, S),$$
$$E_t = \phi_E(E_{t-1}, O_t, \Psi_t, S), \qquad J_t = \phi_J(O_t, S, \Psi_t, E_t),$$

and the state and event records are written back before the next turn is processed. The order follows from what each inference needs before it can be made. The utterance "I'm leaving" marks a collapse of trust in a hostage negotiation and resignation in a farewell scene, so the latent state cannot be inferred before the scene is fixed. Whether "as you wish" is a concession or a passive-aggressive move depends on the speaker's hostility and intent, so causal attribution in the event chain has to wait for the latent state. Judgment comes last because it needs the event chain: a reply may be apt when read alone and still contradict an established fact or close off the course the interaction was taking.

The number of layers is fixed by a different argument, namely by what would be lost if any two were merged. A scene absorbed into the latent state would become revisable from turn to turn, and a judge could then relax a constraint to accommodate the very behavior under assessment. A latent state absorbed into the event chain could register a change in a character only as something that happened, whereas a character who conceals a collapse of trust has changed while nothing visible has happened; judgments of concealment and restraint rest on this difference. An event chain rebuilt by the adjudicator at each turn would accumulate nothing, so failures that develop over many turns would go unrepresented. Finally, if description and adjudication were merged, every revision of the rubric would invalidate the archived records. Table 3 states the resulting design requirements, each with the failure it guards against.

### 3.2 Data layer: constructing the scenario space

The inferences an evaluation licenses extend no further than the situations it samples. When scenarios are collected opportunistically, results can be reported for the scenarios at hand, but one cannot tell which kinds of situation were never visited, nor how far the results generalize. We construct the scenario set by design instead. Following facet theory (Guttman and Greenbaum, 1998) and evidence-centered design (Mislevy et al., 2003), we declare in advance the dimensions along which situations vary and sample their full crossing, so that coverage becomes a verifiable property of the corpus (Figure 4). In this respect the evaluation is holistic in what it samples: every stage of the capability progression is tested across a declared space of situations, and each conclusion can state the region of that space to which it applies.

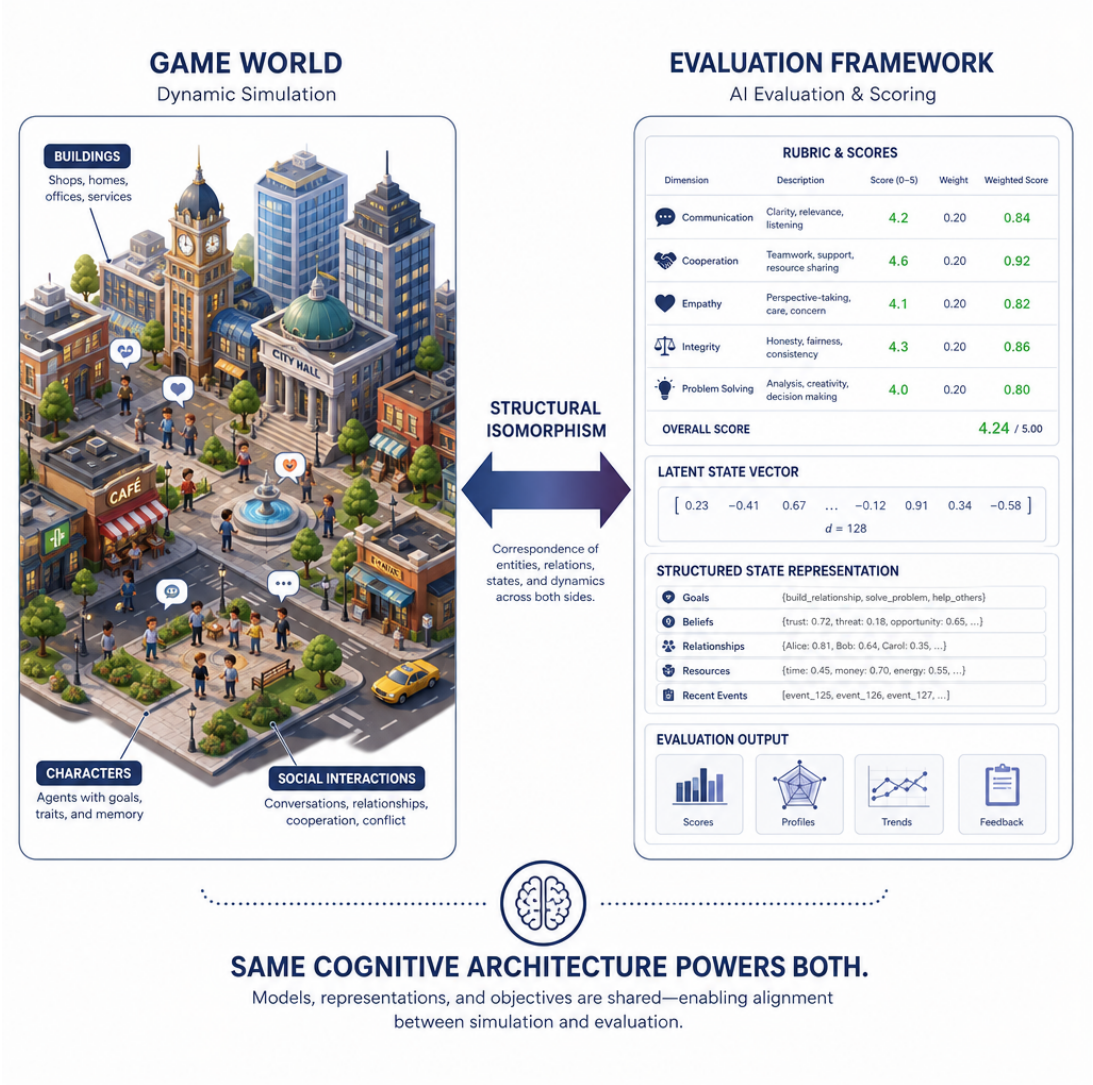



Figure 3: The four-layer architecture read as a symbolic world model: scene constraints, participant states, causal dynamics and value occupy the roles a learned world model would represent, in a form that is inspectable and can be re-scored. Legacy conceptual drawing: dimensional labels do not represent the implemented slot schema or six-level scale and need revision before submission.

Table 3: Design requirements and the failure each prevents.

| Requirement | Operational criterion | Failure if absent |
|---|---|---|
| M1 Cognition-first | Represent the intentions, affect, beliefs, goals and relations needed to explain an utterance | Fluent but psychologically implausible turns pass |
| M2 Temporal continuity | Evaluate turns and trajectories with cross-turn writeback | Locally acceptable, globally degenerating interactions remain invisible |
| M3 Causal grounding | Attach verbatim evidence and explicit transition rationales | A score cannot say where or why failure occurred |
| M4 Hierarchical decomposition | Return an L1–L6 capability vector under dependency constraints | A flat score cannot locate the failed capability |
| M5 Descriptive–normative separation | Three descriptive layers followed by a separate adjudicator | Revising the rubric requires rebuilding the world representation |
| M6 Structured reproducibility | Enforce schemas, cascade ceilings, gate order and writeback outside the model | Free-form judge behavior varies across runs and cannot be audited |
| M7 Evaluation–optimization compatibility | Expose state, evidence, score and future-path fields through one record | Evaluation and training reimplement divergent signals |
| M8 Initiative sensitivity | Distinguish responding within the interaction world from acting on it, conditional on the earlier stages | Fluent interactions that never move forward receive full marks |

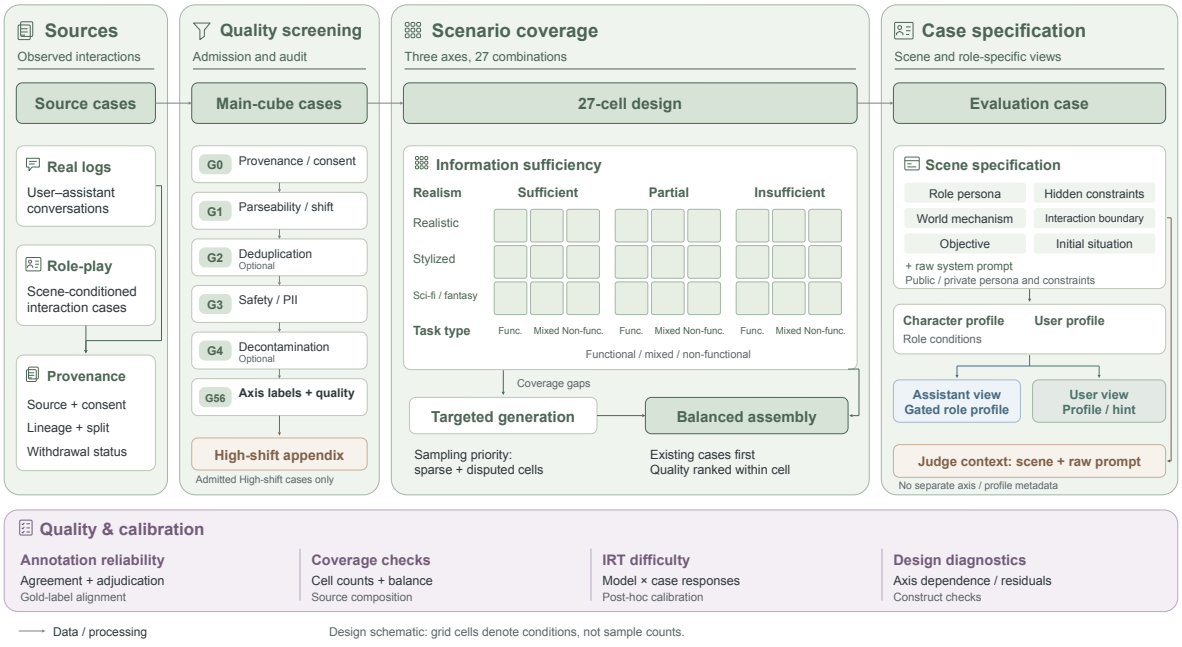



Figure 4: Data-layer design. Source and quality gates feed a three-axis, 27-condition sampling frame and role-specific case views. G2 and G4 require configured components; the cube depicts design conditions rather than measured cell counts. Structured axis and profile metadata are excluded from judge input, although raw scene text may reveal them indirectly.

### 3.2.1 Three facets of a scene

Any specification of an interactive situation has to settle at least three matters: which rules hold in the world, how much of the relevant information the participant is given, and what purpose the interaction serves. We take these ontological, epistemic and teleological questions as the three facets of the scenario space (Table 4). The first, realism, draws on Ryan's principle of minimal departure, which orders fictional worlds by their distance from the actual one; it asks whether a system remains consistent as that distance increases. Information sufficiency rests on the notion of common ground and on Grice's maxim of quantity. An underspecified scene violates the maxim, and a system facing it may proceed on what it has, ask for what is missing, or invent it. The third facet, interaction purpose, takes up Bales's distinction between task-oriented and socio-emotional activity and asks whether a system's competence depends on there being a task to serve. We do not claim that these facets exhaust the properties of a scene. Character traits are carried by the profiles, cooperativeness belongs to the interlocutor, and turn count and modality are parameters of administration. What can be verified is more limited: that the facets are declared in advance, that every cell of the crossing is populated, and that no scored construct varies systematically outside the facets.

Table 4: The three facets of the scenario space.

| Facet | Levels | Construct probed |
|---|---|---|
| A. Realism | `realistic` / `stylized` / `sci_fi` | Consistency under world rules that depart from everyday reality |
| B. Information sufficiency | `sufficient` / `partial` / `insufficient` | Inference under underspecification: filling gaps, seeking clarification, avoiding fabrication |
| C. Interaction purpose | `functional` / `mixed` / `non_functional` | Goal-directed versus relational interaction |

Each facet has three levels, giving 27 cells, and the number is set by the effects the design must be able to detect. A two-level factor has one degree of freedom and supports only a linear trend, so it cannot distinguish a monotone response from a non-monotone one. Non-monotonicity is expected on the information facet. A specification that supplies nothing makes it plain that the system should ask, whereas one that supplies half of what is needed leaves the system to decide whether the gap matters and gives it room to fill the gap on its own; the intermediate level may therefore be the hardest. A center point added to a two-level design would reveal that curvature is present but could not attribute it to a facet, since the quadratic effects of the three facets would be confounded. With three levels each facet has two orthogonal contrasts, and curvature can be both estimated and attributed. The intermediate levels are, in addition, conditions in their own right. A stylized world departs from reality in the entities it contains while keeping its physics, which is a different departure from that of a science-fictional world, and the three levels of information sufficiency call for different correct responses: to proceed, to judge whether to ask, and to ask. Four or five levels would have raised the crossing to 64 or 125 cells, more than our budget of scenes per cell could support.

An earlier version of the design used the interlocutor's cooperativeness as the second facet. We replaced it because cooperativeness depends on which system plays the user, not on the scene, and because it is measured directly in the partner role (§4.5, §4.5.1); stratifying on it would have duplicated that measurement. Information sufficiency is a property of the scene and can be annotated before any dialogue is generated.

### 3.2.2 Constructing the corpus

The present corpus is drawn from two observational sources. Most of it comes from character cards that users of a Chinese role-play platform (造梦) wrote for their own immersive play. These cards already specify hidden motives, counterfactual world mechanisms and behavioral boundaries, the constructs the instrument is designed to probe. The second source, logs of real human–machine dialogue, keeps the corpus close to the way users actually open and sustain interactions. The design provides for three further sources that are not part of the corpus used here: expert-written scenarios with reference trajectories, characters adapted from literary and dramatic works, and red-team seeds. Because adversarial conduct is attributed to the interlocutor and not to the scene, red-team seeds are meant to contribute scenes with sparse information, not hostile ones. Every sample carries a provenance record and is screened for safety and personal information, after which a sequence of filters removes unparseable and duplicated material and, where the required components are configured, material that overlaps with the training data of tested systems. The last filter asks whether each facet can be assigned a definite value. A scene that cannot be placed in the crossing contributes nothing to controlled coverage, and a test of decidability can be applied more consistently than a subjective rating of content. On the first two batches this filter

retained 90% and 64% of samples.

Quality has to be judged within a cell. In role-play material quality and difficulty are closely confounded, since a sparse and disordered scene appears both poor and hard, and an absolute quality threshold would remove disproportionately many scenes from the difficult cells the design most needs. Annotators are told the cell to which a scene has been assigned and rate its quality against other scenes of the same kind, so that quality is judged conditionally on the facets.

An audit of the source pools against the three facets revealed a marked and systematic imbalance. Functional scenes made up about 2.6% of the role-play logs, realism was concentrated at the stylized level, and between 10 and 13 of the 27 cells were empty. Rankings computed on these data alone would have described competence only in fictional, relational settings. We filled the under-populated cells by conditional generation, producing each new scene from its cell specification together with a complete scene description and profiles for both participants. Generated scenes were deduplicated against the existing pool and compared with real scenes in length and field occupancy, a check that exposes the excessive tidiness typical of generated specifications. The resulting balanced set contains 108 scenes, 36 at each level of each facet. A second set of 27 scenes, one per cell, serves the more expensive two-sided experiments.

The difficulty of an item is estimated from how the tested systems respond to it. A difficulty prior recorded at annotation is used only to stratify sampling; the reported difficulty is a parameter fitted to the systems' response patterns (Rasch, 1960). The estimate needs no human responses, and it also guides the selection of future items, which are most informative when their difficulty lies close to the ability of the systems tested.

#### 3.2.3 Participant profiles

Every scene carries a structured profile for each participant, and every field of a profile is grounded in an established psychological construct. The character profile follows the three strata of McAdams and Pals's integrative model of personality, namely dispositional traits, characteristic adaptations and the narrative self, and fills each stratum with its best-replicated taxonomy. Traits are described by the Big Five, motivation by self-determination theory, emotion by the process model of emotion regulation, attachment by Bartholomew's model and interpersonal stance by the agency–communion circumplex. Social cognition is described by the order of theory of mind and by executive control, and the boundaries enforced by the red-line veto by moral foundations. The user profile has the same structure and also records the interlocutor's goal, probing strategy, communicative style and emotional sensitivity. Because the fields are taken from established taxonomies, each inherits the justification that its taxonomy already has, lexical in the case of the Big Five and structural in the case of the circumplex. Trait values are set from the empirical distribution observed in more than 400 real users on 43 dimensions, so that the traits of a persona co-occur as they do in people.

The profile is intended to cover the relevant constructs, not to determine behavior. As Mischel and Shoda argue, dispositions predict behavior only in interaction with situations, which is why the instrument evaluates profile and scene jointly. Background attributes such as name, occupation and history are distal influences on what a character does next. Its next action depends more directly on its appraisal of the situation, its current goals and its model of the partner, and these are what the latent state tracks from turn to turn (§3.4.2) and what the instrument holds a system accountable for.

Facet labels and profiles are used for sampling and for breaking down results, and they are not shown to the judge, which receives the persona, constraints, world mechanism, boundary, objective, initial situation and raw prompt of the scene. This closes the direct route by which the sampling design could influence scores. Raw scene text may still reveal the cell indirectly, and a separate audit is needed to bound that residual. Because profiles are presented to models as text, their serialization is part of the instrument as well; it is fixed and versioned, and its effect on rankings is examined directly (§4.9).

### 3.3 Interaction layer: generating the evaluation trajectory

Once a scene has been drawn, a dialogue has to be produced in it, and the conditions of production constrain what the subsequent evaluation can show. Three conditions matter here: who occupies the two roles, what each participant is permitted to observe, and how generation is kept separate from scoring. They also determine whether initiative can be observed at all. A model can be credited with acting on an interaction only if it cannot see what its partner has withheld and only if the partner's own behavior is measured, and the interaction layer is designed to secure both.

Each episode pairs an assistant policy with a user policy. The user policy is conditioned on the user profile, which sets its goal, probing strategy, communicative style and emotional sensitivity; the assistant

policy is conditioned on the scene and, where the data provide one, on a character profile. The user policy is not regarded as part of the measuring apparatus. Any system may play either role, and the system in the user role is scored on a rubric of comparable depth, so that how persistently it pursues an unresolved point or how cooperatively it behaves is observed and not assumed. We accordingly make no claim that any particular simulated user behaves like a human one. In the one-sided protocol a single reference system plays the user opposite every assistant, holding the partner constant across comparisons. In the two-sided protocol each system plays both roles against every other system, and the partner enters the design as a factor (§4.5).

Asymmetric information between the participants is imposed by the interface. Every persona and hidden-constraint entry in a scene is marked as public or private and assigned to one participant, and private entries are delivered only to their owner. The levels concerned with taking up information and updating state are meant to measure inference from incomplete information, and if the assistant could read the user's private entries they would measure access to hidden information instead.

A related asymmetry concerns information that is available only after the fact. Outside preset role-play, a deployed assistant usually starts without any system profile, so an assistant profile for such data can only be extracted from the completed session, and it is then written with knowledge of every subsequent turn. Such knowledge is acceptable in the user profile, whose function is to represent the user's underlying goal, and a view of the whole session is a reasonable basis for recovering it. In an assistant profile the same knowledge would allow the evaluated system to anticipate the direction of the conversation. We keep the assistant profile for role-play data in which the character has an explicit objective, and omit it for general assistant data.

Dialogues are generated in full before evaluation begins, and nothing produced by the evaluation is returned to the generator. The instrument cannot steer an interaction toward cases that it scores favorably, and an archived dialogue can later be re-evaluated with another judge or a revised rubric. Evaluation proceeds through the completed dialogue one turn at a time, carrying its reconstruction forward, and stops at the turn at which the normative layer finds that the interaction has succeeded or is no longer worth continuing (§3.5.4). How many turns are evaluated is an outcome of the measurement.

### 3.4 Mechanism layer: reconstructing the interaction world

The mechanism layer recovers from a scene and a dialogue the interaction world $\mathcal{W}_{1:T}$ of Section 2. Its three components record what took place and how the interaction developed, and they are kept free of evaluative content, which is left to the normative layer. Figure 5 follows a single turn through them. Where the data layer makes the evaluation holistic in what it samples, the mechanism layer makes it holistic in what it represents: scene, participants' states and the causal course of the interaction are reconstructed together. The event chain in particular supplies the structural evidence on which judgments about the active end of the progression rest.

#### 3.4.1 Scene

At the start of an episode the scene is parsed into six fields: the character's persona, hidden constraints, the world mechanism, the interaction boundary, the objective and the initial situation. Together they describe who is acting (persona and hidden constraints), what governs action (world mechanism and boundary) and what the action is directed toward (objective and initial situation). Each field is also used by a particular part of the evaluation. The persona gives content to the question of whether a character stays in character, and the hidden constraints allow tacit rules and private motives to be tested when the partner probes them. Violations of the world's rules are detected against the world mechanism, and red lines are defined by the boundary. The objective determines what counts as progress in the episode, while the initial situation, which includes power relations and information asymmetries, sets the starting values of the latent state. We expect the same six fields to apply in other domains. In tutoring, for example, the persona corresponds to a teaching style, hidden constraints to learning objectives and the boundary to the limits of permissible help. Whether they suffice outside role-play and assistance has not yet been tested.

Section 2 described the scene as relatively stable, and the instrument treats it as quasi-static: it is carried unchanged through ordinary turns, but the events of a story can legitimately change the world in which it takes place. A character who has gained authority may revise a rule, and a character's development may alter one of its traits. If every attribute were frozen at the outset, the instrument would penalize a character for violating a rule it had itself just changed within the fiction. Revisions are permitted through the event chain alone, and only when the dominant projected path is an advance and the evidence indicates a real change in a trait or rule. A revision amends the existing content and does not replace it, and it is recorded together with its justification, so that later appraisals can be traced to the version of the




Figure 5: Mechanism-layer path for assistant-side evaluation. Scene context conditions the latent-state and event-chain updates, which precede monitoring, the rubric cascade and the ordered halt checks; permitted state carries forward to the next evaluated exchange. The trust-transition and cascade values are schematic rather than measured.

scene they relied on. Projected paths that maintain, degrade or collapse the interaction cannot modify the scene.

### 3.4.2 Latent state

The latent state is the part of the instrument in which the Data Iceberg Theory becomes operational. As argued in §2.1.2, its slots make no claim about a model's private cognition. They are operational variables, admitted only insofar as they help explain observed behavior, constrain what can plausibly follow, and support judgments that can be checked against evidence. The third property separates an explicit state most clearly from holistic judgment. A judge who rates a reply as poor gives no indication of which aspect of the interaction the rating responds to. Against an explicit state, the same judgment can be checked step by step. If a user's deception has just been exposed, for example, the record should show a fall in the character's trust, cite the sentence that exposes the deception, and allow a warm and unguarded reply to be identified as a failure of state progression.

The character's state is described by four families (Figure 6). Internal states, among them trust, vigilance and anxiety, concern how the character feels. Relational states such as intimacy, dominance and boundary violation concern its standing toward the partner. What the character knows of the world is represented by world-mechanism states, including rule awareness and contradiction risk, and what it wants by goal states, which comprise a primary goal, a defensive goal and the current mode of negotiation. Research on social cognition treats these four as the questions a competent agent keeps asking about another (Chen et al., 2024; Shanahan et al., 2023). The same four families result from crossing two distinctions: whether a state concerns the agent itself or what lies outside it, and whether it describes a present condition or is directed toward something. Internal states are conditions of the agent and relational states are directed from it toward the partner; world-mechanism states describe the condition of the world as the agent understands it, and goal states are directed toward outcomes in that world. The candidates we considered adding either fell into one of the four cells, as cognitive load falls among internal states, or spanned two of them, as strategic thinking spans goals and world rules, and would have made the families overlap. A fifth family describes the interaction and not the character. It contains engagement, the value the interaction retains for the other participant, and a provisional indicator of emergence, which marks behavior beyond what the character's profile would predict.

Individual slots were admitted to a family in two steps. Each candidate had first to correspond to a principal dimension in the relevant literature, such as valence, arousal and internal conflict for affect or affiliation and dominance for relations. It then had to meet four empirical conditions: annotators could extract it with acceptable agreement, it explained variance beyond the other slots, the direction of its change could be updated consistently, and removing it altered at least one capability score. Self-esteem, social status and cognitive load did not meet these conditions. Annotators could not distinguish self-esteem reliably from emotional stability, social status hardly varies within a single scene, and cognitive load added nothing beyond emotional stability and internal conflict. An external psychological review led to several further distinctions. Trust was separated from vigilance, since a character may trust an ally's intentions and still be alert to danger, and intimacy was separated from dependency. Dominance was treated as the only bipolar construct in the vocabulary, with a midpoint that denotes parity between the parties and not missing information. Boundary violation was given two components, one tied to the event of transgression and one to its lasting effect.

The value of each slot is one of four ordered degrees, from *very not* to *very*, with no neutral midpoint. It is recorded together with the supporting evidence and reasoning and a short description of the current state. The categorical degree is primary because a model can assign a meaning to "somewhat distrustful" in a way it cannot to 0.37; the numeric value used in computation is a projection of that judgment. Later layers receive only the description. It conveys features of a state that a degree leaves out, such as fluctuation within a turn, and it keeps the numeric values away from the adjudicator, which consequently has nothing numerical to cite as evidence (§3.5.2).

### 3.4.3 State transition

At each turn the latent state is updated from three terms, a carried prior, the new evidence and a modulation derived from the event chain, subject to a significance gate $g_t$:

$$\Psi_t \;=\; \Pi\Big(\alpha\,\Psi_{t-1} \;+\; g_t\,(\Delta_t \odot \mu_t)\Big), \qquad \big\|\Pi(\cdot) - \Psi_{t-1}\big\|_\infty \le \kappa(g_t). \tag{1}$$

The coefficient $\alpha$ represents inertia, $\Delta_t$ the direction of change in each slot as supported by evidence, $\mu_t$ the narrative context, $\Pi$ the projection onto the four degrees, and $\kappa(g_t)$ the largest change permitted by the significance class. The decomposition follows appraisal theory. The prior expresses the tendency of a psychological state to persist when nothing disturbs it, the evidence term corresponds to the novelty and

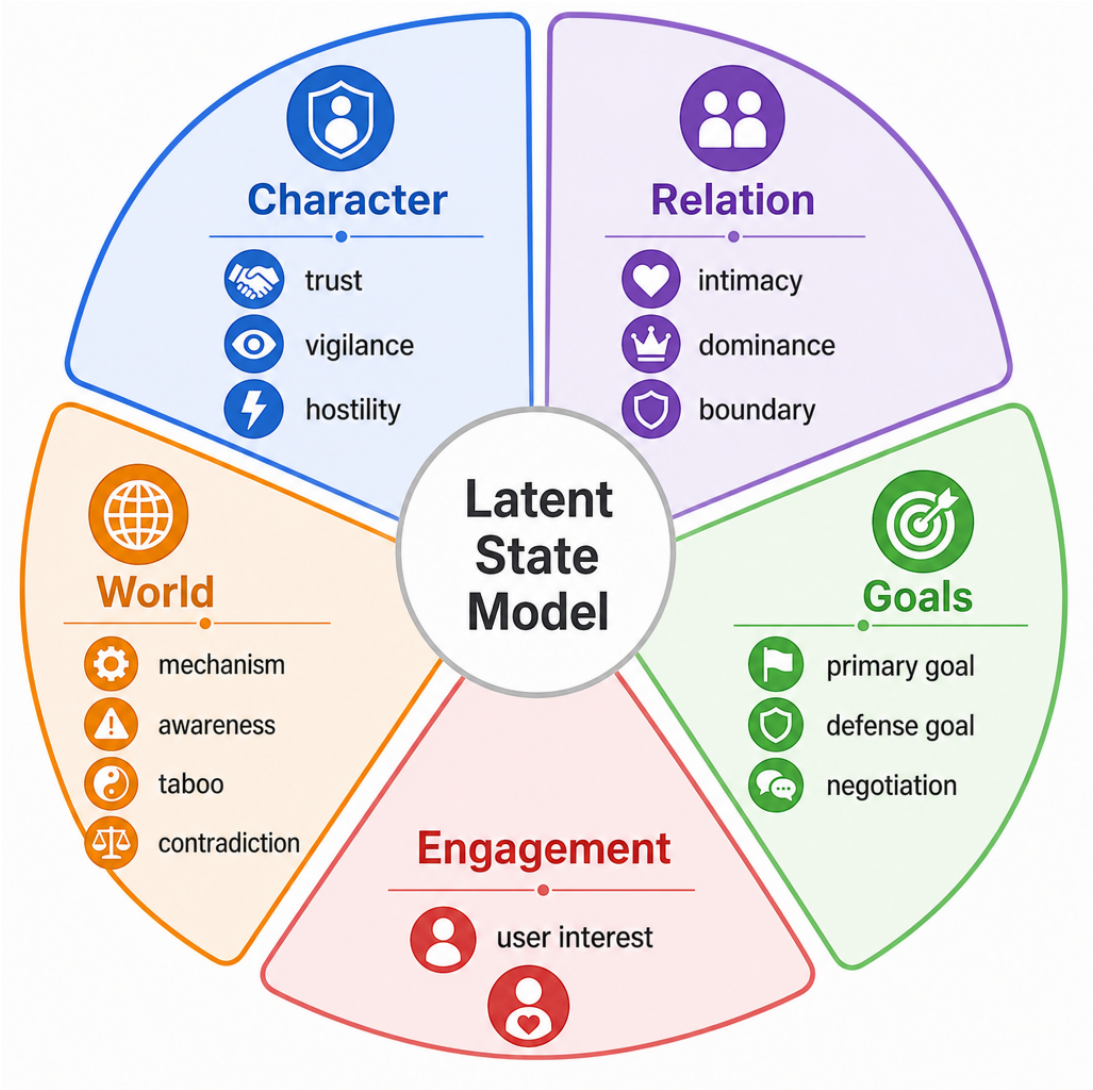



Figure 6: Latent-state families: four describing the character's mind and one describing the interaction.

goal relevance of what has just happened, and the modulation corresponds to coping potential, which is why a betrayal during a high-stakes negotiation shifts the state more than the same betrayal in small talk. Inertia may differ by direction, so that trust can fall quickly and recover slowly. Equation (1) describes the dynamics at the conceptual level only. In practice the model emits ordinal degrees with evidence, and the constraints are enforced outside the model.

Each turn is assigned a significance class before its state is updated. A stable turn leaves the state unchanged, a minor turn allows a change of one degree, and a major turn allows a full reappraisal. Because the class is fixed first, the update can be checked against it: a minor turn that moves a slot by two degrees is inconsistent by definition, and the excess is clamped. The projection keeps a record of changes smaller than one degree. Without it, each turn would be rounded back to the nearest degree, and states that should recover gradually could never recover. The external review pointed out that a turn with no narrative event is not necessarily psychologically inert, since anxiety may subside and dependency may build in the absence of any new stimulus. Such turns are classed as minor, and the small changes they produce accumulate over successive turns until they amount to a change of degree.

Engagement is updated separately and on a shorter timescale, because it concerns whether the interaction is worth continuing and not any state within the story. A character's state can evolve quite plausibly by the standards of the fiction while the exchange loses its value for the participants, so the in-story state cannot by itself show that an interaction should end. Engagement is read on both sides. On the user side it represents interest and is conditioned on the user profile, since an undemanding user may be satisfied by companionship that a demanding user would abandon. On the assistant side it represents the tolerance that the user or the scene retains for the assistant and is conditioned on the scene's requirements for reliability. Because engagement depends on the person as well as on the performance, it is used to decide whether evaluation continues (§3.5.4) and not as a measure of quality.

### 3.4.4 Event chain

Where the latent state operationalizes the Data Iceberg Theory, the event chain operationalizes the Movie Evaluation Theory. It is designed chiefly to capture interactions that are locally coherent but globally degenerative, in which each reply is fluent and appropriate while the interaction as a whole drifts or settles into a loop. Since no individual turn is at fault, the failure is visible only to an evaluator that represents the trajectory. The chain also allows trajectory quality to be assessed directly, for instance by distinguishing a turn that moved the interaction forward from one that only kept it going.

Following §2.2.2, the chain represents $E_t = (P_t, C_t, F_t)$, a past, a present and a future, which correspond to reasoning about what has been observed, about what is currently being acted upon, and about what would follow from a different action (Pearl, 2009). Each of the three regions is further divided into a fixed and an evolving part. For the past, the premise (prior relationships, information asymmetries and the rules of the world) is fixed at the outset; the history grows with each turn. The present comprises the constraints, which define what is permitted, and the active situation, which defines what is at stake. The two must be kept apart because a reply can observe every constraint without engaging what is at stake, or raise the stakes while breaking a rule. For the future, goal tracking concerns the coherence of the participants' intentions and path projection the direction of the interaction. These can diverge: a goal may remain stable while the trajectory decays, and it may change while the trajectory remains healthy. We looked for causally relevant observations in the annotated material that could not be assigned to one of the six components and found none; this supports the partition without proving it complete.

The present is represented at two levels of resolution. The plot point records the objective situation, and attention records what is psychologically salient to the participants, as indicated by the latent state. The two can change independently. A dramatic external event may leave attention where it was, and a pointed remark may shift attention while the situation stays the same. Updates to the chain pass through the present. If either the plot point or attention changes, the past is re-derived as the cause of the new present and the future as its consequence; otherwise the chain is carried over unchanged. Information flows in one direction, from past to present to future, and both large events and small psychological shifts can trigger an update.

The future $F_t = \{f_t^{(1)}, \ldots, f_t^{(K)}\}$ is a set of candidate paths, each classified according to whether it involves structural change and whether its direction is positive. The four resulting categories are exhaustive and mutually exclusive: the interaction may *advance* to a new state, *maintain* its current one, *degrade* by repeating itself, or *collapse*. For each path still considered live, the chain records the reasons it is plausible, its probability, and the conditions under which it could be reached, the last stated in the capability terms used by the rubric. Each projected decline is linked in this way to the capability that, had it been exercised, would have prevented it. The link connects the descriptive and normative parts of the instrument, and it has the counterfactual form that targeted corrective training would require. The projected paths are also the instrument's closest approximation to the quantity $\mathrm{Init}(O_t)$ of Section 2.3: a turn that shifts the

dominant path toward advancement is the structural trace of initiative, and the rubric's assessment of proactive agency (§3.5.1) draws on it.

Participants' goals are often left unstated and have to be inferred. An inferred goal accepted as fact would allow an evaluator to explain any behavior whatever. The chain holds an inferred underlying goal as a hypothesis, recorded with the evidence for and against it, alternative explanations and a confidence level, any of which may be revised later. Behavior is judged consistent with a goal only if the hypothesis has independent support and has withstood the evidence against it; a hypothesis framed to account for the turn under evaluation does not qualify, and hypotheses without adequate support are not passed to the adjudicator.

### 3.5 Normative layer: capability adjudication and halt

The normative layer produces the adjudication $(J, V, h)$ of Section 2.4: capability judgments $J$ (§3.5.1), red-line decisions $V$ (§3.5.3) and the termination decision $h$ (§3.5.4). It reads the scene, the latent state and the event chain and does not modify them. Section 2.4 gave two reasons for this separation, and each has a concrete consequence here. Because the standard is not built into the reconstruction, a revised rubric can be applied to archived records without reconstructing them. Because the evidence for a judgment must come from the interaction record and not from the evaluator's own conclusions, the adjudicator is subject to explicit rules against circular inference (§3.5.2). Where the latent state and the event chain operationalize the first two principles of Section 2, the normative layer operationalizes the third. Its central task is to measure the capability progression of Section 2.3, and in particular to tell a model that responds well from one that acts on the interaction world.

#### 3.5.1 Operationalizing the progression: L1–L6

Social and role-play benchmarks commonly score a flat list of dimensions (Zhou et al., 2024; Tu et al., 2024). Such lists take no account of dependencies among the capabilities they name, and a low engagement score, for example, does not reveal whether the character's internal state has stalled, its expression has become monotone, or it has fallen back on narration. The rubric instead operationalizes the progression of Section 2.3 as six levels ordered by dependency, two for each stage (Table 5). Passive execution is assessed by goal establishment (L1) and information absorption (L2), responsive adaptation by state progression (L3) and expression alignment (L4), and active adaptation by proactive agency (L5) and long-horizon coherence (L6). Within the progression the levels also vary along two dimensions, depth of processing, from receptive through responsive to generative, and temporal range, of which only L6 extends beyond the single turn. In role-play, L1 appears as fidelity to the persona, which is one realization of the level and not its definition.

Table 5: The capability hierarchy. Both roles share one causal spine; the wording of each level is written separately for each role (§4.5).

| Level | Assistant-side criterion | User-side criterion |
|---|---|---|
| L1 | Establish and preserve the operative goal | Establish the user's goal |
| L2 | Absorb new information and its implications | Focus on the relevant part of the assistant's response |
| L3 | Progress the state or task using absorbed information | Preserve the user role rather than drifting into an assistant |
| L4 | Adapt expression to this user and context | Match the specified user profile |
| L5 | Proactively change the interaction in a useful direction | Probe proactively or revise the goal reflectively |
| L6 | Maintain cross-turn goals, state and world coherence | Maintain cross-turn consistency with the user profile |

Each step along the progression presupposes the one before it (Mislevy et al., 2003). A goal gives direction to the uptake of information, and absorbed information is what a state progresses on. Expression is judged against an established state, and proactive moves build on coherent expression, so that a system can act on the interaction world only after it has registered and responded to that world. Long-horizon coherence comes last, since consistency is informative only in a participant that acts. We settled on six levels after considering the alternatives. With L1 and L2 merged, a character that breaks role could no longer be distinguished from one that stays in role but misses what matters, although the two failures have different causes and call for different remedies. With L3 and L4 merged, failures of internal state and of expression would be conflated, and turns that pass one of these levels while failing the other are common in our data. A merger of L5 and L6 would likewise conflate initiative within a turn with consistency across turns. In the other direction, splitting L1 into identity and world rules would produce

two levels with the same dynamics and the same evidence, and a creativity level between L5 and L6 would break the dependency order, since creativity neither requires nor guarantees consistency.

To keep a failure from being attributed to several levels at once, each level specifies what it does not assess. State progression concerns what the character feels or decides, as shown by its behavior, and does not consider wording. Expression alignment concerns only whether the wording conveys that state; a character that deliberately conceals its feelings passes, and a failure is recorded only when the system cannot modulate its expression. Proactive agency is judged by what a turn does in the exchange, so a well-written turn that achieves nothing fails; initiative is credited for its effect on the course of the interaction and not for length or assertiveness. Long-horizon coherence is the one level that spans turns; it can fail through drift or a repeated template in an episode in which L1 passes at every turn.

Since each level builds on the one below, failures are propagated upward, and this is also what makes initiative conditional on the earlier stages: a turn that redirects an interaction whose goal has been lost is capped at the level of that earlier failure. Each level receives a raw score $r_\ell \in \{0, 0.4, 0.6, 1\}$, and the reported vector $s$ is obtained by applying a ceiling that can only decrease,

$$s_\ell = \min(r_\ell, c_\ell), \qquad c_1 = 1, \qquad c_{\ell+1} = \begin{cases} 0, & s_\ell = 0, \\ s_\ell, & 0 < s_\ell < \tau, \\ c_\ell, & s_\ell \geq \tau, \end{cases} \tag{2}$$

where $\tau = 0.6$ is the pass threshold (Figure 7). A hard failure sets all higher levels to zero and a soft failure caps them at its own value, while a passed level imposes no constraint, so scores above a passed level need not decrease. We call the highest level that reaches the threshold the capability *plateau* and the lowest failed level the *blocking dimension*. Both are computed from the capped vector, and the raw scores are kept for inspection. The cascade is applied outside the judge; a judge that capped its own scores would regain the discretion the cascade is meant to remove.

Each level is scored on four points. In a pilot with five points, agreement fell at the lower levels and most disagreements involved adjacent grades, which suggests that annotators could not maintain the finer distinctions (Krippendorff, 2004). With four points the boundary between failure and success is unambiguous, and the two grades below the threshold separate hard from soft failures, which the cascade treats differently. A change from 0.4 to 0.6 then corresponds to passing a capability threshold, an interpretation that differences between adjacent points of a finer scale would not admit.

The wording of the levels was calibrated in three stages. A draft derived from the theory was first piloted. Three senior experts, each responsible for two levels, then scored a larger set blind and resolved disagreements through structured arbitration, and the large-scale review reported in §4.7 came last. Arbitration also produced rules for boundary cases, among them that paraphrasing the partner is necessary but not sufficient for information absorption and that structural repetition counts as a failure of agency and not of expression. In addition, an external panel of psychologists reviewed the constructs, the temporal model, the hierarchy and the termination rules by structured questionnaire, since a construct can be extracted reliably and still be poorly defined.

### 3.5.2 Evidence grounding

For a judgment to count as a measurement, it must be possible to trace it to the interaction. Every level score is accordingly accompanied by quotations from the dialogue, the amount of evidence required increases with the grade, and a grade that lacks citations is reduced. A further risk is circularity. Suppose the latent state records a fall in trust and the adjudicator credits state progression on the strength of that record; the evaluator's own inference would then serve as evidence for itself. For this reason state progression is judged from behavior visible in the dialogue, and the adjudicator receives the latent state only in the form of natural-language descriptions, without the numeric degrees. The same rule applies to inferred goals, which may be cited only with independent support (§3.4.4), and to the facet labels of the sampling design, which are withheld from the judge altogether (§3.2.1). Neither the way scenes were sampled nor the evaluator's intermediate hypotheses can serve as evidence about a system.

### 3.5.3 Red lines

A red line ends an episode immediately, and we keep the set of red lines small. The first kind is a violation of identity or boundary, such as revealing the system prompt, disclaiming in the manner of "as an AI", emitting structured output or template tokens, or breaching a boundary stated by the scene. In each case the system has stepped outside the interaction, and no tolerance is applied. The second kind is mechanical repetition, nearly identical text across turns that indicates a generation loop or a corrupted context. It is distinguished from pattern repetition, in which the wording varies while the same move recurs, and from functional repetition, which serves repair, reassurance or the construction of meaning and is not treated as a fault.

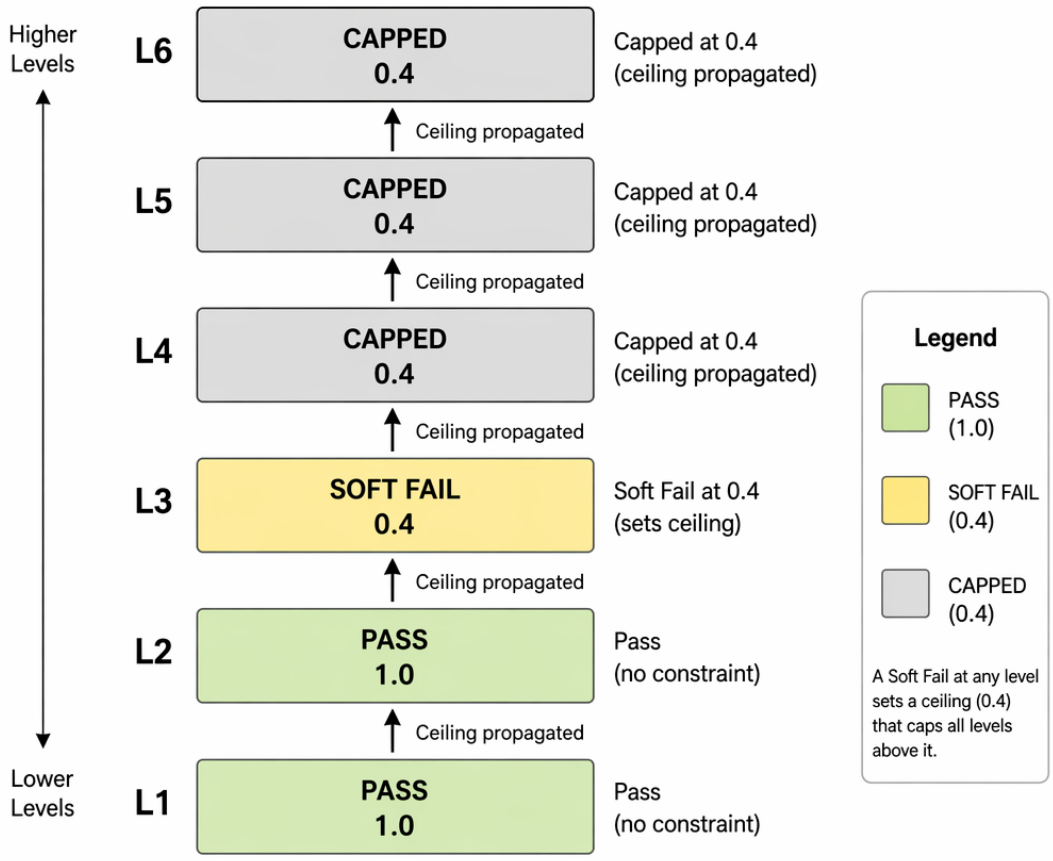



Figure 7: Progressive cascade scoring. A failed level lowers the ceiling for every level above it; the plateau and blocking dimension are read from the capped vector.

### 3.5.4 Halt

Deciding when an interaction should end involves a judgment about its value, and ending every episode after a fixed number of turns leaves that judgment unmade. We recognize two outcomes. An episode succeeds when its objective has been met and further turns would add nothing; it ends by exhaustion of interest when the interaction is no longer worth continuing for the partner. Exhaustion may be abrupt, through a red line (§3.5.3), or gradual, through a decline in engagement.

Behavioral problems that fall short of a red line, including stagnation, forgetting a rule of the world, structural incompleteness, pattern repetition, reversion to templates, depersonalization and loss of direction, do not end an episode directly. They reduce engagement. People leave conversations when they lose interest, and these behaviors are among the reasons they do. How much each problem should cost also depends on the person and the setting: an inattentive user may hardly notice stagnation that a demanding user would find intolerable, and a single lapse matters more in a scene that requires strict reliability than in casual companionship. The decay terms are conditioned on the user profile on the user side and on the scene on the assistant side.

Termination is decided by two coupled quantities,

$$b_t \;=\; b_{t-1} - \big(\beta + \delta_t\big), \qquad \text{halt at } t \text{ if } V_t \text{ fires, } \; v_t \le \theta, \text{ or } b_t \le 0, \tag{3}$$

where $v_t$ is current engagement and $\theta$ its threshold, $b_t$ is a patience budget for the episode, $\beta$ is a baseline cost per turn that is reduced when a turn advances the interaction, $\delta_t$ is the conditioned decay, and $V_t$ is the red-line veto. We set $\theta$ empirically. Episodes were first run with no threshold, and the resulting trajectories were reviewed jointly to identify the point at which a participant would plausibly have stopped. A threshold on current engagement alone has no memory and cannot distinguish a sustained decline from a temporary dip; the patience budget provides the missing memory. It is drawn down at every turn, more heavily after poor turns and less after turns that advance the interaction, and it recovers only slowly. A single poor turn is absorbed by accumulated goodwill, while a long series of mediocre turns can exhaust the budget without any one of them falling below the threshold, which is closer to how a human interlocutor's patience behaves.

An episode is closed as a success when the rubric indicates that the objective has been achieved, the goal has not drifted pathologically, and the dominant projected path is an advance or maintenance. Successful closure is the terminal form of active adaptation: an interaction that has been carried forward, or deliberately held, and then brought to an end. Treating success as a distinct outcome prevents a completed interaction from being scored as an early collapse, and it provides examples of ending an interaction well, a capability that per-turn rewards do not capture. The four conditions are checked in a fixed order: the red-line veto first, then successful closure, then the engagement threshold, and finally the patience budget (Figure 8). A low rubric score does not by itself end an episode.

### 3.5.5 From record to score

Section 2.1.3 treated a scalar score as an aggregation of an evaluation record, and the instrument keeps the two apart. The composite reported here is a weighted mean of the capped level vectors. Higher levels and later turns receive more weight, a turn on which a red line is crossed contributes nothing, and turns that an episode did not reach are counted against it, so that a system whose episodes end early is judged on the whole horizon and not on its easier opening turns alone. We report the composite together with quantities that it does not determine: the stage means for passive execution (L1–L2), responsive adaptation (L3–L4) and active adaptation (L5–L6), the ratio L6/L5, the difference $\Delta T$ between non-functional and functional performance, retention in the hardest cell of the cube, the distribution of terminal states, and the rates of fabrication and clarification. All of them are computed from the same records.

Within each layer, we separate judgment from procedure. Deciding which slots changed, what caused a change and whether the cited evidence supports a grade requires interpreting text, and this is done by a model. Schema validation, enforcement of the step limits, carrying state forward, applying the cascade and ordering the termination checks are carried out by deterministic code, because a model asked to perform them would not perform them identically across runs. Two evaluations of the same transcript can then differ only in the model's appraisals, and the size of that difference can be estimated directly (§4.9).

The record also has the structure of a decomposed reward. The capped level vector, the plateau, the blocking dimension, the projected path and the termination outcome are available through one interface, so the quantities aggregated for a leaderboard are the same ones a training procedure would use, and no separately engineered reward can diverge from the evaluation. The plateau and the blocking dimension name the level at which a policy falls short, which a scalar reward cannot do. Whether optimizing against this record improves interactive competence is an empirical question that this paper leaves open.

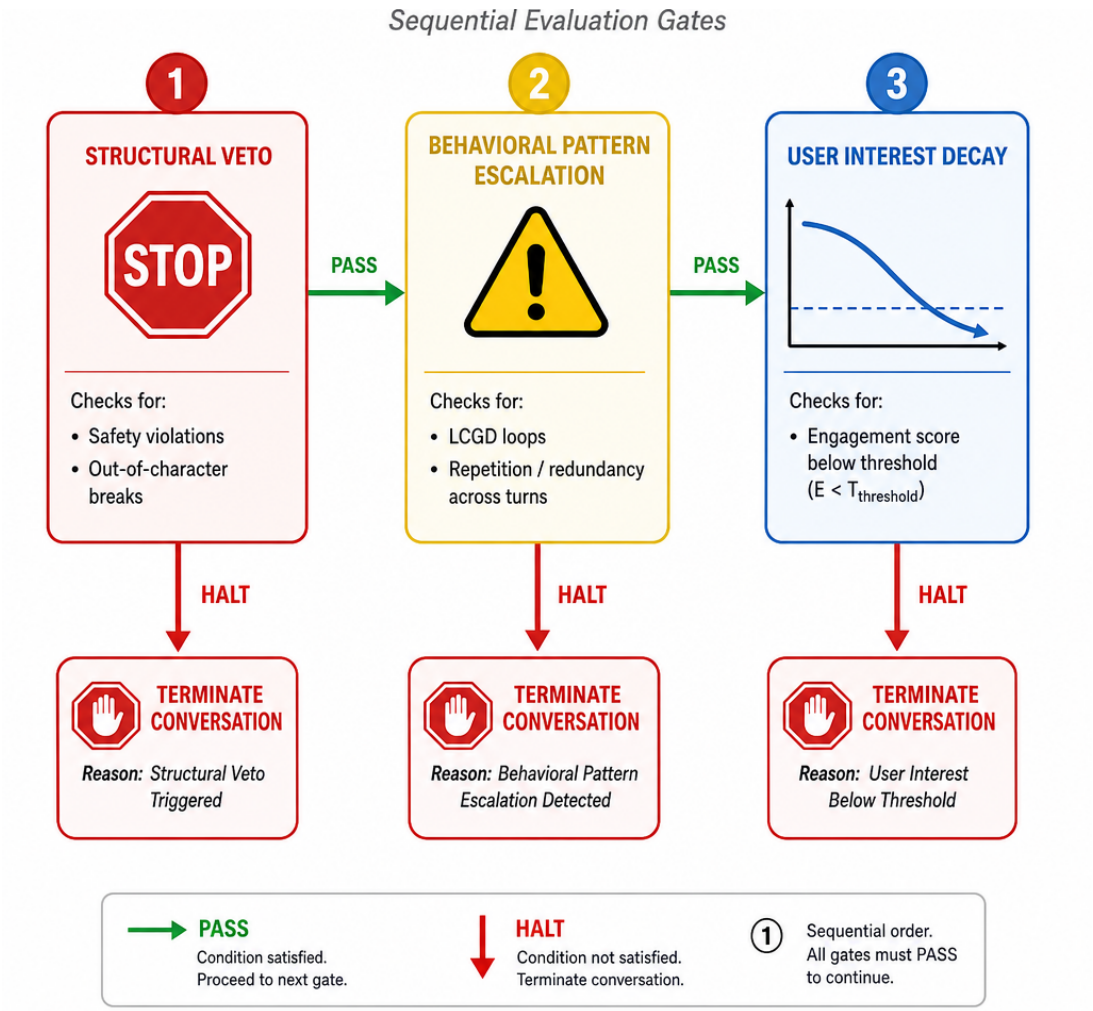



Figure 8: Value-driven halt logic. Legacy drawing: the code-ordered decision is VETO, SUCCESS, low engagement, exhausted patience, then NONE; MAX_TURNS is enforced by the outer loop. A low rubric score alone is not a halt kind.

## 4 Experiments and Results

The theory of Section 2 makes claims that can be checked against data. If the observable surface underdetermines the interaction world, systems that look alike on a single score should differ in how they fail. If local quality does not imply trajectory quality, some failures should be visible only across turns. If initiative is a capability in its own right, differences between systems should concentrate at the active end of the progression and should not be predictable from passive execution alone. And if the partner is part of the interaction, it should also be part of every score. **The experiments ask whether these predictions hold and, where they do, what the instrument reveals about contemporary systems.** We first examine the capability profiles of ten systems and what a composite score conceals about them (§4.2), their response to the designed scenario conditions (§4.3), and two behaviors that no capability score expresses (§4.4). We then score both participants and estimate how much of each score belongs to the partner (§4.5), and extend the protocol to cognitive-architecture agents and to groups of more than two (§4.6). The comparison with human reviewers (§4.7), the analysis of trajectory-level failure (§4.8) and controls on the robustness of the pipeline (§4.9) follow.

### 4.1 Setup

Each system was evaluated on all 27 cells of the cube with thirty dialogues per cell, or 810 episodes per pass. The cube was run three times independently, giving 2,430 episodes per system, and all tabulated values are means over the three passes. Claude Opus 5 served as judge throughout, so judge identity is constant across systems and conditions; its influence on the results is examined in §4.9. In the one-sided protocol, `claude-5-opus` played the user; in the two-sided protocol (§4.5.1) each system played both roles. The judge had no access to facet labels, nor to participant profiles in cases where a deployed system would not have them. Raw scene text was visible, however, and may indicate a cell indirectly. Episodes ended according to the termination rules of §3.5.4. The ten systems are a sample of current frontier models, and we note where a conclusion depends on which systems were included.

### 4.2 The frontier lies at the active end

Table 6: Level profile over the 27-cell cube, means of three passes. The three stage columns are the means of L1–L2 (passive execution), L3–L4 (responsive adaptation) and L5–L6 (active adaptation). $\Delta T$ is non-functional minus functional performance, so a negative value marks a task-leaning system. The final row gives the range across the roster.

| | | level | | | | | | stage | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | final | L1 | L2 | L3 | L4 | L5 | L6 | L1–L2 | L3–L4 | L5–L6 | L6/L5 | $\Delta T$ |
| claude-opus-5.5 | 0.744 | 0.933 | 0.907 | 0.874 | 0.833 | 0.660 | 0.603 | 0.920 | 0.854 | 0.631 | 0.914 | −0.132 |
| gpt-6-astra | 0.736 | 0.927 | 0.899 | 0.865 | 0.824 | 0.655 | 0.596 | 0.913 | 0.845 | 0.625 | 0.910 | −0.124 |
| claude-fable-5.1 | 0.721 | 0.924 | 0.901 | 0.862 | 0.814 | 0.630 | 0.571 | 0.913 | 0.838 | 0.600 | 0.906 | −0.129 |
| claude-5-opus | 0.632 | 0.909 | 0.882 | 0.828 | 0.766 | 0.488 | 0.436 | 0.896 | 0.797 | 0.462 | 0.894 | −0.171 |
| gpt-5.6-sol | 0.602 | 0.917 | 0.880 | 0.811 | 0.746 | 0.441 | 0.391 | 0.899 | 0.779 | 0.416 | 0.887 | −0.014 |
| qwen3.8-max | 0.564 | 0.903 | 0.872 | 0.804 | 0.719 | 0.381 | 0.336 | 0.888 | 0.762 | 0.358 | 0.881 | −0.107 |
| gemini-3.8-flash | 0.526 | 0.903 | 0.871 | 0.786 | 0.702 | 0.318 | 0.275 | 0.887 | 0.744 | 0.297 | 0.865 | +0.004 |
| deepseek-v4-pro | 0.503 | 0.928 | 0.914 | 0.848 | 0.773 | 0.218 | 0.181 | 0.921 | 0.811 | 0.200 | 0.832 | −0.152 |
| kimi-k3 | 0.497 | 0.899 | 0.864 | 0.778 | 0.690 | 0.252 | 0.243 | 0.882 | 0.734 | 0.247 | 0.965 | −0.082 |
| glm-5.3-flash | 0.484 | 0.892 | 0.866 | 0.771 | 0.677 | 0.247 | 0.214 | 0.879 | 0.724 | 0.231 | 0.868 | +0.021 |
| range | 0.260 | 0.041 | 0.050 | 0.103 | 0.156 | 0.442 | 0.422 | 0.042 | 0.130 | 0.431 | 0.133 | 0.192 |

The level profiles in Table 6, which also report the composite, the three stage means, the ratio L6/L5 and the purpose disposition $\Delta T$, show that differences between systems widen stage by stage along the progression. The range across the roster increases monotonically from 0.041 at L1 to 0.442 at L5 and remains of similar size at L6 (0.422). Averaged by stage, passive execution spans 0.042, responsive adaptation 0.130 and active adaptation 0.431. For these systems and this corpus, passive execution is close to saturation; its levels mainly locate where an interaction breaks down and do little to separate the systems from one another. **Current systems differ above all in how well they carry an exchange forward and keep it consistent over many turns**, far less in whether they hold a goal and take in what they are told. A rubric that weighted all six levels equally would devote most of its resolution to the levels on which the systems are almost indistinguishable.

The stages also measure different things. `deepseek-v4-pro` and `claude-opus-5.5` have the strongest passive execution on the roster and are practically indistinguishable there (0.921 and 0.920), yet their

active adaptation differs by a factor of three (0.200 and 0.631). `gpt-5.6-sol`, weaker in passive execution (0.899), is about twice as strong in active adaptation as `deepseek-v4-pro` (0.416). Reliable goal maintenance and information uptake do not imply initiative or consistency across turns, and a single latent scale would hide the difference. The same analysis led us to drop a measure of our own. An index summarizing the two levels of active adaptation, which we had intended to report, improves fit by only 0.0002 once L5 is included, since under ordinary conditions it is a rescaling of L5.

For every system L6/L5 is below one, and for nine of the ten it lies between 0.832 and 0.914; ordinarily, then, long-horizon coherence amounts to a roughly constant discount on proactive agency. `kimi-k3` departs from this pattern with a ratio of 0.965, losing 0.009 between the two levels where the other systems lose between 0.033 and 0.059. A long effective context would produce this pattern, since fewer contradictions of established facts mean a smaller penalty on the only level that spans turns. The cascade does not force the ratio below one, because L6 can exceed L5 whenever L5 passes, so the regularity is empirical.

#### 4.2.1 What a composite score conceals

Three systems lie within 0.019 of each other on the composite. We do not consider differences this small to be resolvable and present the three systems without ranking them. Their profiles, however, differ considerably (Table 7). Within the same 0.019 band, $\Delta T$ ranges over 0.173 and L6/L5 over 0.132. `deepseek-v4-pro` pairs passive execution as strong as any on the roster with the weakest active adaptation, `kimi-k3` shows the smallest drop from L5 to L6 of any system, and `glm-5.3-flash` is the only one of the three that performs better in non-functional than in functional settings. This is the case anticipated in §2.1.3, in which two trajectories satisfy $q(\tau_1) \approx q(\tau_2)$ while differing in their failure modes. **For these three systems the composite is accurate as an aggregate and uninformative as a diagnosis.** The argument for reporting profiles holds regardless of which of the three is judged the best.

Table 7: Diagnostic columns for the three systems whose composite scores are indistinguishable. *Retention* is the proportion of episodes surviving in the hardest cell of the cube.

| | final | L1–L2 | L3–L4 | L5–L6 | L6/L5 | $\Delta T$ | retention |
|---|---|---|---|---|---|---|---|
| deepseek-v4-pro | 0.503 | 0.921 | 0.811 | 0.200 | 0.832 | −0.152 | 0.703 |
| kimi-k3 | 0.497 | 0.882 | 0.734 | 0.247 | 0.965 | −0.082 | 0.753 |
| glm-5.3-flash | 0.484 | 0.879 | 0.724 | 0.231 | 0.868 | +0.021 | 0.827 |
| span | 0.019 | 0.042 | 0.087 | 0.048 | 0.132 | 0.173 | 0.124 |

Retention in the hardest cell, in which the world departs from real rules, information is insufficient and the purpose is non-functional, is not predicted by the composite. `gpt-5.6-sol`, fifth on the composite, retains 0.841 of its episodes in this cell, more than `gpt-6-astra` or `claude-fable-5.1`, while `claude-5-opus`, fourth on the composite, retains 0.659. Retention is instead associated with $\Delta T$. Systems whose functional and non-functional scores are close retain the most episodes, and among the three systems in Table 7 retention increases as $\Delta T$ approaches and passes zero. The design suggests an explanation. The hardest cell provides neither a clear task nor sufficient information, so the strengths measured at the passive and responsive stages have little to work on, and the systems that hold up are those whose competence does not depend on an instrumental framing of the interaction.

### 4.3 The scenario facets have distinct effects

A full crossing of three facets earns its cost if the facets affect behavior in different ways, and Table 8 shows that they do. Realism behaves as a difficulty dimension, with every system scoring lower as the setting moves further from everyday plausibility. Interaction purpose shows no common pattern. Seven systems score highest in functional and lowest in non-functional settings, one scores highest in non-functional settings, and two score highest at the mixed level, where instrumental and socio-emotional demands arise together and have to be balanced. $\Delta T$ in Table 6 summarizes this variation and bears almost no relation to overall strength: the two systems closest to zero are seventh and tenth on the composite, and the most task-leaning system is fourth. It is therefore reported beside the composite as a separate column.

**The half-given specification.** The effect of the information facet has to be estimated with the scenario held constant, since pooling across scenarios mixes the effect of withholding information with differences among the scenarios from which it was withheld. Estimated within each model over matched scenarios, the effect is not monotone (Table 9). All three systems examined score lowest at the intermediate level: **the hardest specification is the one that is half given, not the one that is empty**. This is the possibility that motivated three levels in the design (§3.2.1). When a specification is empty, it is plain that the system

Table 8: Scores by level of the purpose and realism facets, means of three passes. Realism runs from everyday settings through stylized ones to settings that deviate from real-world rules.

| | interaction purpose | | | realism | | |
|---|---|---|---|---|---|---|
| Model | functional | mixed | non-functional | everyday | stylized | deviant |
| claude-opus-5.5 | 0.795 | 0.774 | 0.663 | 0.784 | 0.747 | 0.701 |
| gpt-6-astra | 0.785 | 0.763 | 0.661 | 0.776 | 0.739 | 0.694 |
| claude-fable-5.1 | 0.774 | 0.745 | 0.645 | 0.762 | 0.720 | 0.682 |
| claude-5-opus | 0.702 | 0.664 | 0.531 | 0.678 | 0.643 | 0.575 |
| gpt-5.6-sol | 0.604 | 0.612 | 0.590 | 0.642 | 0.600 | 0.564 |
| qwen3.8-max | 0.601 | 0.598 | 0.494 | 0.619 | 0.582 | 0.493 |
| gemini-3.8-flash | 0.515 | 0.544 | 0.519 | 0.575 | 0.533 | 0.471 |
| deepseek-v4-pro | 0.569 | 0.524 | 0.417 | 0.540 | 0.498 | 0.473 |
| kimi-k3 | 0.530 | 0.514 | 0.448 | 0.536 | 0.492 | 0.463 |
| glm-5.3-flash | 0.471 | 0.491 | 0.492 | 0.527 | 0.485 | 0.441 |

should ask; when it is half complete, the system has to judge whether the gap matters and may be tempted to fill it on its own. A design that compared only sufficient and insufficient specifications would have found a monotone effect and would have located the difficulty at the insufficient level. The intermediate level also shows about six times the within-scenario variance of the other two, which indicates where our specification of the facet is least complete.

Table 9: The information facet, estimated within model over matched scenarios. The middle panel gives within-scenario variance by level; the lower panel splits the intermediate level into three variants: the gap flagged explicitly, the gap concealed, and the gap concealed together with a false premise that conflicts with the rest of the specification.

| Model | sufficient | partial | insufficient |
|---|---|---|---|
| gpt-6-astra | 0.774 | **0.692** | 0.705 |
| gpt-5.6-sol | 0.652 | **0.545** | 0.566 |
| gemini-3.8-flash | 0.565 | **0.460** | 0.515 |
| within-scenario variance | 0.00143 | **0.01046** | 0.00208 |
| | explicit gap | concealed gap | concealed + false premise |
| gpt-6-astra | 0.701 | 0.717 | **0.657** |
| gpt-5.6-sol | 0.582 | 0.582 | **0.472** |
| gemini-3.8-flash | 0.502 | 0.466 | **0.412** |

We therefore split the intermediate level into three variants. Concealing the gap has little effect, and the strongest system scores slightly higher when the gap is concealed than when it is flagged. Adding a false premise that conflicts with the rest of the specification lowers every system's score by between 0.04 and 0.11. The intermediate level had combined two conditions, an incomplete brief and an incorrect one, and the systems handle the first considerably better than the second. The result bears on the design of the instrument as much as on the systems, and it motivates a revision in which incomplete and incorrect specifications are treated as separate levels.

### 4.4 Behaviors that capability scores do not express

At every turn we also record two behaviors that no capability score reflects directly: whether the system supplies a fact that the specification does not contain, and whether it asks for one (Table 10).

Across this roster, **fabrication at an information gap rises with competence in passive execution**. The four systems strongest in passive execution, `deepseek-v4-pro`, `claude-opus-5.5`, `gpt-6-astra` and `claude-fable-5.1`, are also the four that fabricate most, and the two lowest rates belong to systems in the lower half of the ranking, contrary to the expectation that invented detail indicates weak capability. The rate tracks passive execution more closely than the composite: `deepseek-v4-pro` is first in passive execution, eighth on the composite and fourth in fabrication. The clearest exception is `gpt-5.6-sol`, fifth in passive execution and third lowest in fabrication. One possible explanation is that a system with a firm grasp of the situation is more willing to infer what is missing than to ask for it, and that a task-oriented disposition puts a premium on producing an actionable answer, whereas asking a question costs a relationally oriented system little. The two behaviors also depend differently on the information facet. Clarification increases steadily as more information is withheld. Fabrication peaks instead when

Table 10: Fabrication and clarification rates, means of three passes, beside each system's passive-execution score (L1–L2) and composite. The lower panel gives the rates by level of the information facet.

| Model | L1–L2 | final | fabrication | clarification |
|---|---|---|---|---|
| claude-opus-5.5 | 0.920 | 0.744 | 0.168 | 0.219 |
| gpt-6-astra | 0.913 | 0.736 | 0.174 | 0.208 |
| claude-fable-5.1 | 0.913 | 0.721 | 0.175 | 0.224 |
| claude-5-opus | 0.896 | 0.632 | 0.139 | 0.177 |
| gpt-5.6-sol | 0.899 | 0.602 | 0.065 | 0.234 |
| qwen3.8-max | 0.888 | 0.564 | 0.111 | 0.189 |
| gemini-3.8-flash | 0.887 | 0.526 | 0.035 | 0.246 |
| deepseek-v4-pro | 0.921 | 0.503 | 0.149 | 0.172 |
| kimi-k3 | 0.882 | 0.497 | 0.078 | 0.217 |
| glm-5.3-flash | 0.879 | 0.484 | 0.027 | 0.274 |

| Level of the information facet | fabrication | clarification |
|---|---|---|
| sufficient | 0.051 | 0.070 |
| partial | **0.164** | 0.203 |
| insufficient | 0.103 | **0.374** |

the specification is half complete, the same level at which scores are lowest in Table 9. Fabrication at an information gap should consequently not be read as an inverse indicator of capability, and it needs to be measured separately from the capability scores.

**How episodes end.** Since episodes end according to the termination rules and not after a fixed number of turns, the manner in which they end is itself informative (Table 11). An episode ends either in successful closure, when the objective has been met and further turns would add nothing, or in exhaustion of interest, which covers a gradual loss of engagement, depletion of the patience budget and abrupt termination at a red line.

Table 11: Distribution of terminal states, and successful closure by interaction purpose, means of three passes. *func.* and *non-func.* give the rate of successful closure in functional and in non-functional settings.

| Model | success | interest exhausted | func. | non-func. |
|---|---|---|---|---|
| claude-opus-5.5 | 12.4% | 87.6% | 23.1% | 3.8% |
| gpt-6-astra | 10.5% | 89.5% | 20.6% | 3.5% |
| claude-fable-5.1 | 10.5% | 89.5% | 20.6% | 3.6% |
| claude-5-opus | 11.8% | 88.2% | 21.9% | 3.4% |
| gpt-5.6-sol | 3.2% | 96.8% | 7.8% | 1.8% |
| qwen3.8-max | 9.1% | 90.9% | 14.2% | 2.7% |
| gemini-3.8-flash | 2.0% | 98.0% | 4.3% | 0.2% |
| deepseek-v4-pro | 9.4% | 90.6% | 16.1% | 2.8% |
| kimi-k3 | 5.4% | 94.6% | 11.2% | 1.3% |
| glm-5.3-flash | 2.0% | 98.0% | 1.8% | 0.2% |

For every system, successful closure accounts for a minority of episodes, so the fact that an episode ended does not in itself count against a system. Nor does the rate of successful closure follow capability. **Successful closure reflects the shape of the task at least as much as the competence of the system**: all systems close successfully several times more often in functional than in non-functional settings, and for `gemini-3.8-flash` the ratio exceeds twenty. The rate is comparable only within a level of the purpose facet. Pooled over the whole cube, it penalizes systems with a relational disposition for not closing objectives that non-functional scenes do not pose, and we advise against using it as a headline metric in that form.

### 4.5 Two-sided evaluation

When only one participant in an interaction is scored, the behavior of the other becomes an unmeasured component of every score. Simulated interlocutors that appear plausible have been shown to distort the ranking of the systems they are used to evaluate (Dou et al., 2025), and InteractEval accordingly scores the user side as well. Both roles are assessed on the same six levels with the same cascade. The wording of each level is written separately for each role, and the two rubrics are kept as separate documents so that a judge assessing one side is not influenced by definitions intended for the other (Table 5).

The user-side levels have content of their own. Establishing a goal at L1 means, for the user, knowing what it is asking for, and absorbing information at L2 means noticing which part of the assistant's reply calls for a response, for instance an error, a flaw in reasoning or an evasion. L3 on the user side corresponds to staying in character: the user has to keep to its role and not drift into advising, summarizing or the register of a helper. At L4 the user must conform to its assigned profile. The user's version of L5 is either to press its own goal or to revise it after reflection, and the second is the more demanding. L6 requires consistency across turns on both sides. The latent state, the event chain, the red-line scan and successful closure all describe the character's world and are applied only on the assistant side.

With both roles scored, a scenario cube produces a matrix of results, and the same cube shows how well each system performs its role and how demanding it is as a partner. We use the $10 \times 10$ crossing to examine whether the partner acts only as a difficulty setting or interacts with the system being scored, what properties make a partner informative, and how systems fare when paired with themselves. The crossing also provides an empirical check on the choice of scenario facets, since cooperativeness, which we decided not to use as a facet, appears here as a measured property of each system in the partner role.

#### 4.5.1 The partner as a factor: the interaction matrix

In Table 12 every system appears in both roles. Rows correspond to the system playing the partner and columns to the system being scored, and diagonal cells are self-play. Reading down a column shows how a system performs against the whole roster, and reading along a row shows how a system affects the scores of the systems it partners. When `claude-5-opus` plays the partner, the resulting row is identical to the one-sided leaderboard of Table 6; one-sided evaluation is a special case of the two-sided protocol.

Table 12: Interaction matrix over the full $10 \times 10$ roster, means of three passes. Rows are the system playing the partner, columns the system being scored (`o5.5` = `claude-opus-5.5`, `dsk` = `deepseek-v4-pro`). Diagonal cells ($\diamond$) are self-play. The `claude-5-opus` row ($\dagger$) is the standard-user condition and equals the leaderboard of Table 6. *mean* and *span* are the row mean and the max–min across the ten scored columns.

| partner \ scored | o5.5 | astra | fable | opus | sol | qwen | gemini | dsk | kimi | glm | mean | span |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| claude-opus-5.5 | 0.701$\diamond$ | 0.803 | 0.638 | 0.628 | 0.537 | 0.548 | 0.397 | 0.466 | 0.526 | 0.417 | 0.566 | 0.406 |
| gpt-6-astra | 0.775 | 0.792$\diamond$ | 0.786 | 0.702 | 0.499 | 0.631 | 0.397 | 0.579 | 0.526 | 0.322 | 0.601 | 0.470 |
| claude-fable-5.1 | 0.684 | 0.681 | 0.669$\diamond$ | 0.585 | 0.557 | 0.512 | 0.467 | 0.441 | 0.447 | 0.427 | 0.547 | 0.257 |
| claude-5-opus$^\dagger$ | 0.744 | 0.736 | 0.721 | 0.632$\diamond$ | 0.602 | 0.564 | 0.526 | 0.503 | 0.497 | 0.484 | 0.601 | 0.260 |
| gpt-5.6-sol | 0.640 | 0.664 | 0.651 | 0.564 | 0.535$\diamond$ | 0.484 | 0.458 | 0.430 | 0.428 | 0.415 | 0.527 | 0.249 |
| qwen3.8-max | 0.707 | 0.788 | 0.779 | 0.691 | 0.601 | 0.617$\diamond$ | 0.499 | 0.555 | 0.546 | 0.423 | 0.621 | 0.365 |
| gemini-3.8-flash | 0.552 | 0.596 | 0.592 | 0.492 | 0.480 | 0.424 | 0.384$\diamond$ | 0.364 | 0.352 | 0.351 | 0.459 | 0.245 |
| deepseek-v4-pro | 0.714 | 0.774 | 0.776 | 0.726 | 0.461 | 0.597 | 0.343 | 0.586$\diamond$ | 0.475 | 0.278 | 0.573 | **0.498** |
| kimi-k3 | 0.633 | 0.777 | 0.759 | 0.675 | 0.636 | 0.611 | 0.519 | 0.544 | 0.534$\diamond$ | 0.433 | 0.612 | 0.344 |
| glm-5.3-flash | 0.574 | 0.621 | 0.608 | 0.522 | 0.491 | 0.450 | 0.418 | 0.386 | 0.374 | 0.378$\diamond$ | 0.482 | 0.247 |
| column mean | 0.672 | 0.723 | 0.698 | 0.622 | 0.540 | 0.544 | 0.441 | 0.485 | 0.470 | 0.393 | 0.559 | — |

Were the partner no more than a difficulty setting, each row would be a rescaled copy of the column means, and every cell would be determined by its row and column margins. In fact the ordering of the scored systems changes from one partner to another, and 10.9% of the total sum of squares remains after row and column main effects are removed. We had decided in advance to treat pairing as a factor in its own right if this proportion exceeded one-tenth. **The ability to perform a role well and the tendency to press a partner hard are separable**, and a benchmark with a single fixed partner reports the combination of the two as though it were a property of the scored system alone.

This has consequences for how an evaluation partner should be chosen, and the row mean is not a suitable basis for the choice. A partner can lower all scores without separating the systems. `gemini-3.8-flash` yields the lowest row mean (0.459), yet the ten scored systems differ by only 0.245 against it, while `deepseek-v4-pro` yields the widest spread (0.498) at much higher absolute scores. Across the ten partners, spread correlates with probing intensity at $+0.48$, and the spread is what carries information about the systems being scored. Table 13 locates each partner's effect relative to the standard user. Partners that probe hard reduce scores at the active stage (L5–L6), where advancement and consistency are assessed, and partners with low cooperativeness reduce scores at L2–L3, where uptake of the partner's intent is assessed. `claude-fable-5.1` and `gpt-5.6-sol` have similar row means (0.547 and 0.527) but act on different parts of the progression: the first mainly on the active stage ($+0.056$ relative to L2–L3), the second mainly on L2–L3 ($-0.104$). Selecting a partner by the average shortfall it produces may yield one that provides too little usable information, and the low scores it produces then say more about the partner than about the systems being evaluated.

Cooperativeness in these tables is measured from each system's turns in the partner role. Its values

Table 13: Where a partner's pressure falls, relative to the standard user (`claude-5-opus`), means of three passes. ΔL2–L3 and ΔL5–L6 are the additional shortfall induced at the levels of uptake (L2–L3) and of active adaptation (L5–L6) beyond the standard user (positive means those levels are pressed harder), so `claude-5-opus` is the zero baseline; the last column is their difference.

| Partner | probing | coop. | ΔL2–L3 | ΔL5–L6 | difference |
|---|---|---|---|---|---|
| claude-opus-5.5 | 0.112 | 0.87 | −0.012 | +0.047 | +0.059 |
| gpt-6-astra | 0.128 | 0.88 | −0.044 | +0.040 | +0.084 |
| claude-fable-5.1 | 0.118 | 0.90 | −0.004 | +0.052 | +0.056 |
| claude-5-opus | 0.082 | 0.86 | 0.000 | 0.000 | 0.000 |
| gpt-5.6-sol | 0.058 | 0.80 | +0.103 | −0.001 | −0.104 |
| qwen3.8-max | 0.030 | 0.69 | +0.113 | −0.065 | −0.178 |
| gemini-3.8-flash | 0.014 | 0.61 | **+0.291** | −0.025 | −0.316 |
| deepseek-v4-pro | 0.088 | 0.80 | +0.048 | +0.008 | −0.040 |
| kimi-k3 | 0.046 | 0.76 | +0.061 | −0.045 | −0.106 |
| glm-5.3-flash | 0.034 | 0.72 | +0.190 | −0.008 | −0.198 |

fall between 0.61 and 0.90, and none of the ten systems behaves as an adversarial user, consistent with the observational corpus, in which nearly all exchanges are collaborative. These measurements support the decision to use information sufficiency as the second facet. Stratifying a third of the cube by cooperativeness would have meant designing around a property that belongs to one participant, that the user-side rubric already measures, and that varies within a narrow range in practice.

**Self-play.** Table 14 compares each system's score when paired with itself, taken from the diagonal of the matrix, with its score against the standard user. Whether the self-play score is higher or lower is associated with self-similarity, defined as the agreement between a system's probing style as partner and its disposition as character. The four most self-similar systems, `gpt-6-astra` (0.99), `deepseek-v4-pro` (0.90), `kimi-k3` (0.83) and `qwen3.8-max` (0.81), score higher against themselves than against the standard user, the less self-similar systems score lower, and the correlation between self-similarity and the net gap is −0.65. The row and column effects of the matrix lead to the same conclusion by a different route: diagonal cells exceed the values these effects predict by +0.024 on average, compared with −0.003 for off-diagonal cells. Systems tend to be scored more favorably when the partner shares their style, and the advantage grows with the resemblance between a system's two roles and not with its overall strength, to which self-similarity is only weakly related (−0.30). The two analyses agree in direction, and with ten diagonal cells the size of the effect can be estimated only approximately. The estimate is an upper bound, because systems from the same family are stylistically closer than systems from different families, and the bias is better controlled by balancing pairs for stylistic distance than by adjusting scores afterwards.

Table 14: Self-play, from the diagonal of Table 12, beside the score against the standard user. Self-similarity is the agreement between a system's probing style as partner and its disposition as character; *net gap* is the standard-user score minus the self-play score, so a negative value marks a system that scores higher against itself.

| Model | self-similarity | self-play | standard user | net gap |
|---|---|---|---|---|
| claude-opus-5.5 | 0.47 | 0.701 | 0.744 | +0.043 |
| gpt-6-astra | 0.99 | 0.792 | 0.736 | −0.056 |
| claude-fable-5.1 | 0.42 | 0.669 | 0.721 | +0.052 |
| claude-5-opus | 0.43 | 0.632 | 0.632 | 0.000 |
| gpt-5.6-sol | 0.67 | 0.535 | 0.602 | +0.067 |
| qwen3.8-max | 0.81 | 0.617 | 0.564 | −0.053 |
| gemini-3.8-flash | 0.56 | 0.384 | 0.526 | +0.142 |
| deepseek-v4-pro | 0.90 | 0.586 | 0.503 | −0.083 |
| kimi-k3 | 0.83 | 0.534 | 0.497 | −0.037 |
| glm-5.3-flash | 0.65 | 0.378 | 0.484 | +0.106 |

**The user role.** The same crossing scores the partner as well. Applying the user-side rubric to each partner's own turns, with the same roster, cube and protocol, gives each system's profile in the user role (Table 15) and its user-side composite against every assistant (Table 16). The capability progression reappears on the user side. Its stages span 0.066, 0.084 and 0.225 across the roster, so systems in the user role, as in the assistant role, differ most at the active end, in whether they probe and revise their goals (L5) and stay consistent with their profile across turns (L6). User-side L5 also correlates at +0.94 with the probing intensity measured independently from the same turns in Table 13, which is evidence that the

user-side rubric registers what it is meant to. The role has a failure of its own as well. `gpt-6-astra` and `deepseek-v4-pro` score lowest on user-side L3 (0.792 and 0.763), the level that records a user drifting into advising, summarizing and the register of a helper; one plausible reading is that systems trained mainly as assistants carry that register into a role that should not have it. The most consequential user-side result concerns attribution. Of the variance in Table 16, 94.4% is attributable to the system in the user role, 3.6% to the assistant and 2.0% to the pairing, against 10.9% for pairing on the assistant side. How well a system plays the user is almost entirely a property of that system; the assistant's score, by contrast, depends materially on which system plays the user. User-side and assistant-side composites are correlated across systems (+0.88), but the orderings differ: `claude-fable-5.1` is the strongest user and third as an assistant, and `claude-5-opus`, the standard user of the one-sided protocol, ranks fourth in the user role. Figure 9 shows an earlier four-by-four pilot covering both roles; because it used a different roster, it is not pooled with the present results.

Table 15: User-side capability profile: each system's scores in the user role, averaged over the ten assistants it partners and over three passes. The stage columns are the means of L1–L2, L3–L4 and L5–L6 under the user-side wording of Table 5. The final row gives the range across the roster.

| | | level (user side) | | | | | | stage | | |
|---|---|---|---|---|---|---|---|---|---|---|
| System in user role | final | L1 | L2 | L3 | L4 | L5 | L6 | L1–L2 | L3–L4 | L5–L6 |
| claude-fable-5.1 | 0.731 | 0.951 | 0.902 | 0.861 | 0.842 | 0.611 | 0.598 | 0.927 | 0.852 | 0.605 |
| claude-opus-5.5 | 0.724 | 0.949 | 0.905 | 0.853 | 0.838 | 0.604 | 0.586 | 0.927 | 0.846 | 0.595 |
| gpt-6-astra | 0.715 | 0.947 | 0.911 | 0.792 | 0.821 | 0.628 | 0.574 | 0.929 | 0.807 | 0.601 |
| claude-5-opus | 0.707 | 0.944 | 0.889 | 0.858 | 0.829 | 0.562 | 0.571 | 0.917 | 0.844 | 0.567 |
| qwen3.8-max | 0.653 | 0.931 | 0.862 | 0.824 | 0.794 | 0.468 | 0.512 | 0.897 | 0.809 | 0.490 |
| kimi-k3 | 0.643 | 0.928 | 0.857 | 0.831 | 0.781 | 0.452 | 0.497 | 0.893 | 0.806 | 0.475 |
| deepseek-v4-pro | 0.637 | 0.936 | 0.874 | 0.763 | 0.772 | 0.509 | 0.463 | 0.905 | 0.768 | 0.486 |
| gpt-5.6-sol | 0.628 | 0.929 | 0.861 | 0.806 | 0.776 | 0.441 | 0.468 | 0.895 | 0.791 | 0.455 |
| gemini-3.8-flash | 0.585 | 0.912 | 0.823 | 0.817 | 0.748 | 0.352 | 0.421 | 0.868 | 0.783 | 0.386 |
| glm-5.3-flash | 0.577 | 0.907 | 0.818 | 0.809 | 0.741 | 0.368 | 0.392 | 0.863 | 0.775 | 0.380 |
| range | 0.154 | 0.044 | 0.093 | 0.098 | 0.101 | 0.276 | 0.206 | 0.066 | 0.084 | 0.225 |

Table 16: User-side interaction matrix, means of three passes: each cell is the user-side composite of the row system when paired with the column system as assistant; the diagonal (◇) is self-play and *mean* is the row mean. Labels as in Table 12.

| user \ assistant | o5.5 | astra | fable | opus | sol | qwen | gemini | dsk | kimi | glm | mean |
|---|---|---|---|---|---|---|---|---|---|---|---|
| claude-opus-5.5 | 0.724◇ | 0.737 | 0.736 | 0.734 | 0.724 | 0.718 | 0.698 | 0.722 | 0.713 | 0.735 | 0.724 |
| gpt-6-astra | 0.718 | 0.733◇ | 0.745 | 0.705 | 0.712 | 0.708 | 0.696 | 0.706 | 0.705 | 0.721 | 0.715 |
| claude-fable-5.1 | 0.732 | 0.751 | 0.756◇ | 0.735 | 0.728 | 0.744 | 0.714 | 0.718 | 0.722 | 0.710 | 0.731 |
| claude-5-opus | 0.725 | 0.706 | 0.729 | 0.714◇ | 0.693 | 0.707 | 0.708 | 0.693 | 0.693 | 0.703 | 0.707 |
| gpt-5.6-sol | 0.631 | 0.656 | 0.656 | 0.625 | 0.615◇ | 0.634 | 0.617 | 0.615 | 0.614 | 0.616 | 0.628 |
| qwen3.8-max | 0.661 | 0.691 | 0.654 | 0.651 | 0.646 | 0.661◇ | 0.644 | 0.640 | 0.639 | 0.644 | 0.653 |
| gemini-3.8-flash | 0.600 | 0.586 | 0.602 | 0.588 | 0.581 | 0.583 | 0.581◇ | 0.575 | 0.581 | 0.574 | 0.585 |
| deepseek-v4-pro | 0.658 | 0.644 | 0.649 | 0.636 | 0.642 | 0.631 | 0.624 | 0.624◇ | 0.621 | 0.643 | 0.637 |
| kimi-k3 | 0.665 | 0.656 | 0.648 | 0.639 | 0.648 | 0.646 | 0.640 | 0.621 | 0.635◇ | 0.632 | 0.643 |
| glm-5.3-flash | 0.597 | 0.593 | 0.595 | 0.577 | 0.581 | 0.580 | 0.564 | 0.574 | 0.560 | 0.554◇ | 0.577 |
| column mean | 0.671 | 0.675 | 0.677 | 0.660 | 0.657 | 0.661 | 0.649 | 0.649 | 0.648 | 0.653 | 0.660 |

### 4.6 Extensions of the protocol

The protocol applies without change to agents built on an explicit cognitive architecture and to groups of more than two participants, and both have been evaluated on the full cube. A third extension, an indicator of emergent behavior, was run over the full cube and returned a null result.

**Cognitive-architecture agents.** An agent that embeds a base model in an explicit cognitive architecture is evaluated in the same way as the base model, so the protocol can test whether the architecture adds interactive competence and where. We built two such agents on each of the ten systems, METAMIND (Zhang et al., 2026b) and COGNIZER, both of which add structured reasoning about the partner's mental state, and evaluated all twenty agents on the full cube (Table 17). Before the runs we had predicted that an effective architecture would show its gain at the active stage, since frontier systems are already close to ceiling in passive execution. The prediction holds without exception: for each of the twenty agents




Figure 9: Assistant–user interaction in an earlier four-by-four pilot, shown for context and kept separate from the ten-system matrix of Table 12: (a) overall trust by user simulator (row) and evaluated assistant (column); (b) assistant- and user-context means by role; (c) illustrative scenario examples. The pilot uses a different roster and is neither pooled with nor a check on the $10 \times 10$ results.

the gain is largest in active adaptation and smallest in passive execution. Averaged over the ten bases, METAMIND adds 0.014 in passive execution, 0.030 in responsive adaptation and 0.051 in active adaptation, and COGNIZER adds 0.014, 0.045 and 0.078 (Table 18). The gain grows stage by stage, mirroring the spans across systems in Table 6, and it is largest where the base is weakest: the increase in active adaptation falls almost linearly with the base system's own level ($r = -0.99$ for METAMIND and $-1.00$ for COGNIZER), from 0.029 and 0.048 on `claude-opus-5.5` to 0.073 and 0.107 on `deepseek-v4-pro`. A relation this close is likely to owe part of its strength to headroom, since a base already near the top of the scale has less room to rise, and the ten bases do not allow that effect to be separated from a genuine dependence of the gain on the base system's weakness. Neither architecture closes the gap between systems. With COGNIZER, `deepseek-v4-pro` reaches 0.307 in active adaptation, about half of what `claude-opus-5.5` achieves without any architecture.

The two architectures also move initiative in different directions. METAMIND raises $\Delta T$ for every one of the ten bases, by 0.030 on average, and gains most in non-functional settings (0.054, against 0.025 in functional ones). COGNIZER lowers $\Delta T$ for every base, by 0.034 on average, and gains most in functional settings (0.071, against 0.037). A composite score records both as improvements of the same kind, of 0.039 and 0.059. The profile shows that the first architecture helps most where the interaction is relational and the second where it is instrumental, so two interventions that raise initiative by similar amounts push it in opposite directions along the purpose facet. This is the distinction between the amount and the direction of initiative drawn in Section 2.3, observed in an intervention rather than in a comparison across systems.

**Groups beyond the dyad.** The adjudicator scores each participant on the same levels whatever the number of participants, so the protocol extends to groups of three or more systems. A group interaction differs from a dyad in that each turn must be judged against several partners, and the latent state and event chain must represent several relationships and goals at once. We first held the other members fixed, letting the standard user play every other seat (Table 19). Moving from two to four participants lowers the passive and responsive levels (L1–L4) by 0.034 on average and active adaptation by 0.123, more than three times as much, and long-horizon coherence suffers most: the mean ratio L6/L5 falls from 0.892 to 0.595. The decline is steep for systems that were already weak at the active stage and shallow for those that were strong. The three leading systems keep 85–87% of their active adaptation in a four-person group; `deepseek-v4-pro`, `kimi-k3` and `glm-5.3-flash` keep between a quarter and two fifths of theirs; across the roster, the share retained correlates with the dyadic level at $+0.97$. For `deepseek-v4-pro`, L6/L5

Table 17: Cognitive-architecture agents built on each of the ten systems: composite, active adaptation (L5–L6) and purpose disposition $\Delta T$ for the base model, with METAMIND (MM) and with COGNIZER (Cog), means of three passes over the full cube.

| | final | | | L5–L6 | | | $\Delta T$ | | |
|---|---|---|---|---|---|---|---|---|---|
| Base system | base | +MM | +Cog | base | +MM | +Cog | base | +MM | +Cog |
| claude-opus-5.5 | 0.744 | 0.768 | 0.781 | 0.631 | 0.660 | 0.679 | −0.132 | −0.113 | −0.153 |
| gpt-6-astra | 0.736 | 0.762 | 0.775 | 0.625 | 0.653 | 0.670 | −0.124 | −0.113 | −0.137 |
| claude-fable-5.1 | 0.721 | 0.751 | 0.759 | 0.600 | 0.635 | 0.647 | −0.129 | −0.104 | −0.155 |
| claude-5-opus | 0.632 | 0.668 | 0.687 | 0.462 | 0.507 | 0.532 | −0.171 | −0.144 | −0.209 |
| gpt-5.6-sol | 0.602 | 0.638 | 0.662 | 0.416 | 0.463 | 0.493 | −0.014 | +0.010 | −0.044 |
| qwen3.8-max | 0.564 | 0.607 | 0.629 | 0.358 | 0.412 | 0.444 | −0.107 | −0.079 | −0.144 |
| gemini-3.8-flash | 0.526 | 0.573 | 0.595 | 0.297 | 0.359 | 0.388 | +0.004 | +0.035 | −0.035 |
| deepseek-v4-pro | 0.503 | 0.553 | 0.577 | 0.200 | 0.273 | 0.307 | −0.152 | −0.099 | −0.202 |
| kimi-k3 | 0.497 | 0.544 | 0.571 | 0.247 | 0.314 | 0.351 | −0.082 | −0.042 | −0.130 |
| glm-5.3-flash | 0.484 | 0.534 | 0.558 | 0.231 | 0.303 | 0.334 | +0.021 | +0.060 | −0.016 |

Table 18: Mean gain from each architecture over its base model, averaged across the ten base systems: by stage of the capability progression, in $\Delta T$, and by level of the interaction-purpose facet.

| | | stage | | | | interaction purpose | | |
|---|---|---|---|---|---|---|---|---|
| Architecture | final | L1–L2 | L3–L4 | L5–L6 | $\Delta T$ | functional | mixed | non-functional |
| METAMIND | +0.039 | +0.014 | +0.030 | +0.051 | +0.030 | +0.025 | +0.037 | +0.054 |
| COGNIZER | +0.059 | +0.014 | +0.045 | +0.078 | −0.034 | +0.071 | +0.066 | +0.037 |

falls to 0.031, so that in a four-person group its long-horizon coherence all but disappears. Groups widen the differences that dyads already show, and they widen them at the active end: the range of active adaptation across systems grows from 0.431 to 0.498, and that of L6/L5 from 0.133 to 0.819.

Groups of mixed composition show the same trend in the pairing effect (Table 20). The share of variance left after removing the main effects of all members rises from 10.9% in dyads to 13.9% in balanced triads and 16.2% in tetrads, so the larger the group, the less a member's score can be attributed to that member alone. Triads whose members come from the same model family reach 15.1%, somewhat above balanced triads of the same size (13.9%), in line with the self-play result of §4.5.1, where stylistic resemblance between the two roles was associated with a larger joint component. The difference rests on 40 same-family triads and is not large enough to be treated as more than consistent with that result.

Table 19: Group size and the capability progression. Each system is scored as one member of a group whose other members are all played by the standard user (`claude-5-opus`). L1–L4 combines passive execution and responsive adaptation, as the group runs report them jointly. The final row gives the mean over the ten systems.

| | L1–L4 | | | L5–L6 | | | L6/L5 | | |
|---|---|---|---|---|---|---|---|---|---|
| System | 2 | 3 | 4 | 2 | 3 | 4 | 2 | 3 | 4 |
| claude-opus-5.5 | 0.887 | 0.873 | 0.864 | 0.631 | 0.581 | 0.548 | 0.914 | 0.888 | 0.836 |
| gpt-6-astra | 0.879 | 0.866 | 0.856 | 0.625 | 0.576 | 0.538 | 0.910 | 0.899 | 0.827 |
| claude-fable-5.1 | 0.875 | 0.858 | 0.855 | 0.600 | 0.544 | 0.512 | 0.906 | 0.884 | 0.850 |
| claude-5-opus | 0.846 | 0.824 | 0.813 | 0.462 | 0.395 | 0.349 | 0.894 | 0.827 | 0.736 |
| gpt-5.6-sol | 0.838 | 0.814 | 0.806 | 0.416 | 0.343 | 0.293 | 0.887 | 0.822 | 0.726 |
| qwen3.8-max | 0.824 | 0.800 | 0.787 | 0.358 | 0.283 | 0.223 | 0.881 | 0.772 | 0.634 |
| gemini-3.8-flash | 0.816 | 0.794 | 0.774 | 0.297 | 0.212 | 0.154 | 0.865 | 0.707 | 0.548 |
| deepseek-v4-pro | 0.866 | 0.841 | 0.824 | 0.200 | 0.111 | 0.050 | 0.832 | 0.500 | 0.031 |
| kimi-k3 | 0.808 | 0.780 | 0.765 | 0.247 | 0.158 | 0.093 | 0.965 | 0.806 | 0.508 |
| glm-5.3-flash | 0.802 | 0.772 | 0.754 | 0.231 | 0.137 | 0.078 | 0.868 | 0.635 | 0.258 |
| mean | 0.844 | 0.822 | 0.810 | 0.407 | 0.334 | 0.284 | 0.892 | 0.774 | 0.595 |

**Emergent behavior.** The interaction-level family of the latent state includes an indicator for behavior that goes beyond the character's profile, that is, turns showing more insight than the scene, the partner and the profile would lead one to expect, as opposed to competent performance of the role. No capability level rewards such behavior, so it is examined separately. Every turn of every episode was screened for

Table 20: Groups of mixed composition, with members drawn in rotation from the ten systems. The interaction residual is the share of variance left after the main effects of all members are removed; the dyad row is the matrix of Table 12.

| Configuration | group size | groups | mean final | interaction residual |
|---|---|---|---|---|
| Dyad | 2 | 100 | 0.559 | 10.9% |
| Triad, balanced sampling | 3 | 120 | 0.512 | 13.9% |
| Triad, same-family members | 3 | 40 | 0.538 | 15.1% |
| Tetrad, balanced sampling | 4 | 90 | 0.481 | 16.2% |

candidates, with any candidate to be adjudicated by hand. Across 24,300 episodes, 2,430 for each of the ten systems, the screen flagged no turn at all (Table 21). We report this as a null result with two possible readings, between which the present data cannot decide. Contemporary systems may not produce behavior that outruns what their scene, partner and profile would lead one to expect, or the indicator may be too conservative to register it. The second reading is the more cautious one, and a test of it would require seeding the corpus with turns known to exceed the scripted profile and checking that the screen recovers them.

Table 21: Emergent behavior: turns flagged by the screen as exceeding the scripted profile, and the number confirmed on manual adjudication, over the full cube and three passes.

| Systems | episodes | candidate turns | flagged rate | confirmed |
|---|---|---|---|---|
| Each of the ten systems | 2,430 | 0 | 0.000 | 0 |
| All systems | 24,300 | 0 | 0.000 | 0 |

### 4.7 Human review of the evaluation record

All of the preceding results depend on the record produced by the engine. Section 2.1.2 required that latent variables be treated as measurement variables only if their reliability can be assessed, and the same holds for the causal structure and the capability judgments built on them. We therefore asked trained reviewers to examine the record component by component and level by level. The study covered 311 turn-level records from five complete sessions involving four systems, reviewed by 22 annotators (Figure 10). Reviewers audited the engine's structured output instead of scoring the dialogue independently, accepting or rejecting each component and giving a written reason. We adopted the audit design after an earlier round of independent scoring proved uninformative, with 88.8% of items receiving full marks. Reviewers saw the complete record, including the latent snapshot and the event chain, so that they and the engine worked from the same information and disagreements reflect differences of judgment. The review followed the engine's order of operations: the significance decision, the two state updates if the significance decision was endorsed, the six levels, and the red-line decision. The resulting figures are endorsement rates under audit and should not be read as blind agreement coefficients. They were also collected under an earlier version of the rubric (§6.4).

Table 22: Reviewer endorsement of the engine's output, by component and by level. Review of the two state updates is conditional on the significance decision being endorsed.

| Component | endorsed | *N* |
|---|---|---|
| Event-chain update | 99.6% | 262 |
| Red-line determination | 97.4% | 311 |
| Latent-state update | 84.4% | 262 |
| Significance decision | 83.6% | 311 |
| L1 goal establishment | 98.8% | 246 |
| L2 information absorption | 95.1% | 247 |
| L3 state progression | 94.0% | 248 |
| L4 expression alignment | 92.3% | 247 |
| L5 proactive agency | 78.9% | 246 |
| L6 long-horizon coherence | 74.0% | 246 |

Endorsement rates vary considerably across components (Table 22, Figure 11). Reviewers endorsed 99.6% of event-chain updates and 97.4% of red-line decisions. Latent-state updates were endorsed about fifteen points less often, and the rejections followed one main pattern: non-zero values on negative slots in calm




Figure 10: Expert-review protocol.

scenes, and rationales asserting an increase where the recorded degree had not changed. Both are errors in the calibration of the default state, and a stricter default corrects them; they do not suggest that the constructs are unreadable from text. Across the capability levels, endorsement falls steadily from 98.8% at L1 to 74.0% at L6, following the increase in processing depth and temporal range. The engine is least secure where assessment is hardest. The gradient mirrors the result of §4.2: the levels at the active end, which separate contemporary systems most sharply, are also the levels on which human and engine agree least, and judgments about them depend most on the explicit trajectory evidence that the event chain provides.

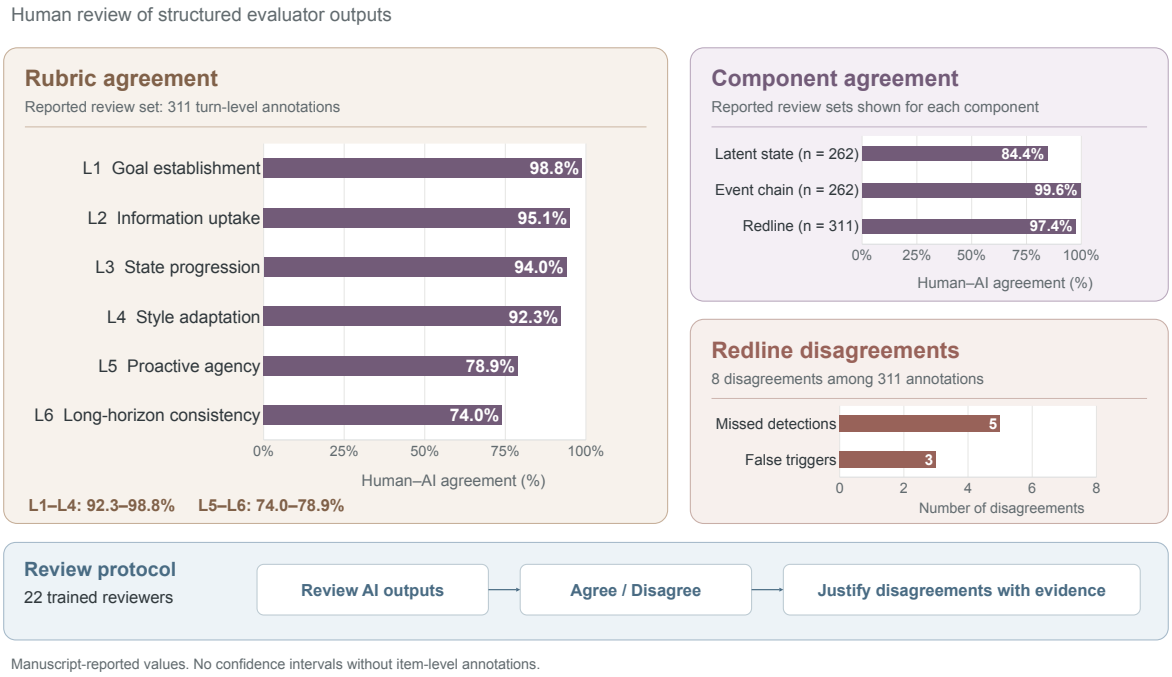



Figure 11: Historical reviewer endorsement by component and level. Twenty-two reviewers considered 311 annotation records, with 262 eligible update reviews and 246–248 reviews per rubric level. The figure predates the current L1 wording and degree ladder; these are output-visible endorsement rates, not blind agreement or current-schema reliability estimates.

The disagreements are concentrated at the active stage, and those coded for direction run almost entirely one way. Counting rejected level ratings instead of turns, the rates in Table 22 correspond to about 165 rejections, some 116 of them at L5 and L6 and fewer than fifty across L1 to L4. Thirty-three rejections were coded for the direction of the error, and in 31 of these the reviewer considered the engine's score too low; the other rejections were not coded, so this asymmetry holds for the coded cases only. The engine's own verdict makes the pattern clearer. When the engine passed all six levels, reviewers endorsed 87.3% of cases ($N = 118$), whereas none of the six cases in which the engine placed the plateau at L2 or below was endorsed. Reviewers and engine disagree mainly about failures.

A review of individual cases attributes most of this disagreement to two reviewer tendencies. Some reviewers awarded the top grade to performance that was competent but not exceptional; the rubric reserves that grade for exceptional performance. Reviewers also tended to take an unchanged goal as evidence of long-horizon coherence, and so credited a character that repeated the same demand in different words over eleven turns with a capability that the trajectory did not show. In one class of cases the reviewers were right and the specification was changed. When a character has left the scene and responds in interior monologue, a requirement to answer the partner's last utterance directly does not apply, yet the engine had applied it and the cascade had propagated the resulting failure upward. Such non-conversational narrative situations are now covered by an explicit rule. Reviewers also differed widely in strictness, endorsing between 29.2% and 96.0% of their items, and did so to a similar degree under both protocols, which suggests that reviewer calibration and not the protocol needs attention.

### 4.8 Failures visible only at the level of the trajectory

In the trajectory analysis we repeatedly observed episodes in which every turn satisfies the passive and responsive levels while projected advancement declines and structural repetition increases (Figure 12).

Each turn considered on its own is acceptable, and the problem appears only in the sequence. The instrument detects such episodes because the rubric and the event chain operate at different temporal scales, whereas an evaluator that computes $q_t = f(O_t, O_{<t})$ has no variable in which the failure could register. Such episodes are instances of the relation stated in §2.2.1,

$$\text{Local Quality} \not\Rightarrow \text{Trajectory Quality},$$

and we take it as support for the Movie Evaluation Theory. It does not show how the behavior arises. Per-turn preference optimization, which rewards safe and agreeable replies even when they change nothing, would produce it, and so would a weakening influence of early goals as a session grows longer; distinguishing the two would require controlled interventions on training and on context.

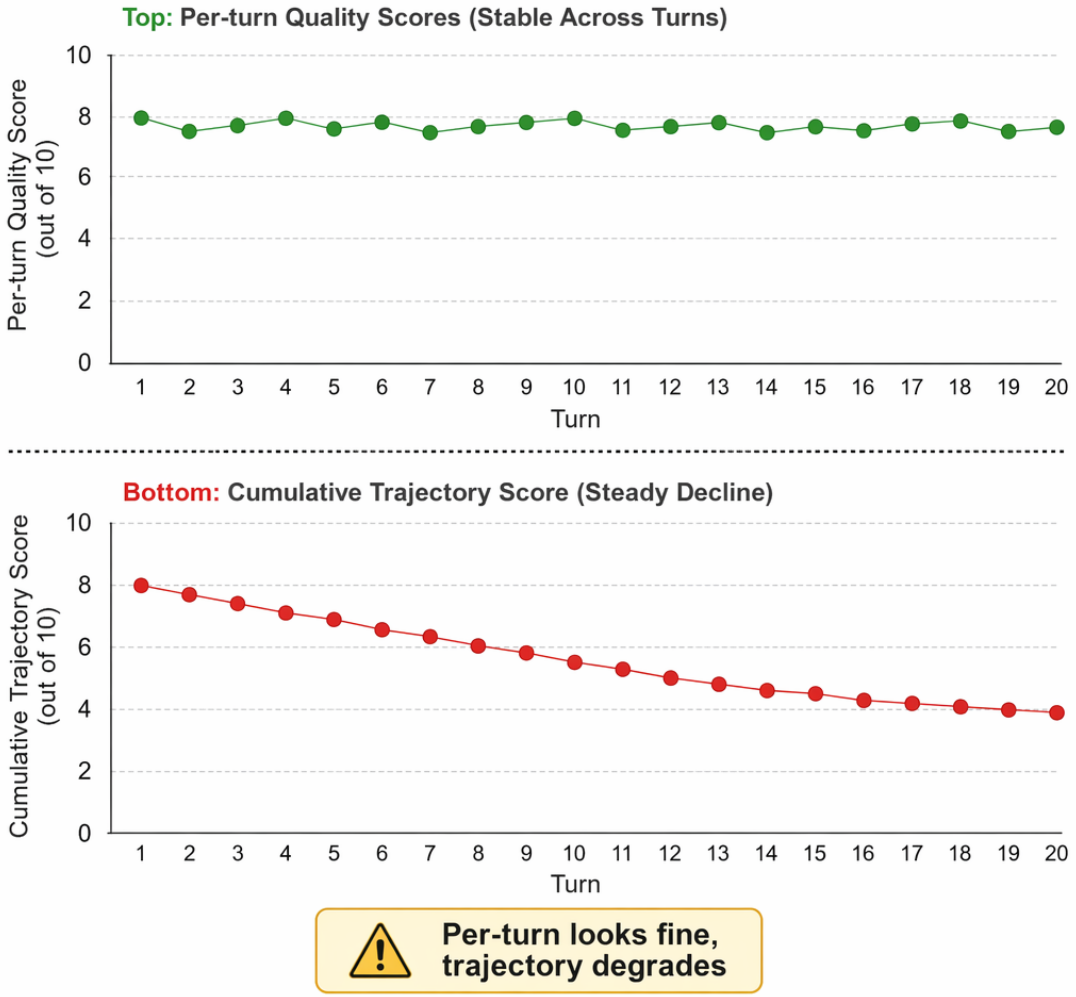



Figure 12: Locally coherent, globally degenerative interaction pattern. Legacy schematic: its plotted points lack episode-level source traces, and its acronym and axis definitions need correction before submission.

The reviewers' disagreements at the active stage are consistent with this pattern. Reviewers were more lenient than the rubric mainly in cases where a reply remained warm and varied in style while repeating the same interactional move, which would be expected if momentary pleasantness were being taken as progress. Kahneman's distinction between experience as it occurs and experience as it is remembered offers one interpretation. Testing it would require separating online from retrospective judgment, and we report the pattern only as a tendency in these annotations. For the design of evaluation the conclusion is more limited: the pleasantness of an exchange and its structural progress should be scored separately, since a single score can be high on the first while the second is absent.

### 4.9 Robustness of the scoring pipeline

All results above come from one configuration of the pipeline, with a single judge, a single model extracting the descriptive records and a single serialization of the input. The controls below each vary

one of these choices and hold the others fixed. The quantity of interest is whether the ordering of systems is preserved; changes in absolute scores are of secondary concern. None of the four controls changes the ordering by more than one rank, and every change that does occur involves systems the reference configuration separates by less than 0.016.

**Test–retest.** All values in this section are means over three independent passes through the cube, and Table 23 reports the passes separately. Every system occupies the same rank in all three, Spearman's $\rho$ between any two passes is 1.000, and the mean absolute change in the composite between passes is 0.003 to 0.004. The largest standard deviation across passes for any system is 0.0036, and the smallest gap between systems adjacent in the ordering is 0.006, between `deepseek-v4-pro` and `kimi-k3`, so even the closest pair is separated by more than the run-to-run variation of either member. This does not make the three systems within 0.019 of each other distinguishable on the composite in any substantive sense (§4.2.1); it means that their order is reproducible, which is a weaker claim.

Table 23: Test–retest reliability over three independent passes of the full cube: composite per pass, mean, sample standard deviation, and the rank each system occupies in the three passes.

| System | pass 1 | pass 2 | pass 3 | mean | SD | ranks |
|---|---|---|---|---|---|---|
| claude-opus-5.5 | 0.743 | 0.743 | 0.746 | 0.744 | 0.0017 | 1 / 1 / 1 |
| gpt-6-astra | 0.738 | 0.734 | 0.736 | 0.736 | 0.0020 | 2 / 2 / 2 |
| claude-fable-5.1 | 0.720 | 0.725 | 0.718 | 0.721 | 0.0036 | 3 / 3 / 3 |
| claude-5-opus | 0.635 | 0.632 | 0.629 | 0.632 | 0.0030 | 4 / 4 / 4 |
| gpt-5.6-sol | 0.599 | 0.606 | 0.601 | 0.602 | 0.0036 | 5 / 5 / 5 |
| qwen3.8-max | 0.561 | 0.564 | 0.567 | 0.564 | 0.0030 | 6 / 6 / 6 |
| gemini-3.8-flash | 0.524 | 0.527 | 0.527 | 0.526 | 0.0017 | 7 / 7 / 7 |
| deepseek-v4-pro | 0.503 | 0.506 | 0.500 | 0.503 | 0.0030 | 8 / 8 / 8 |
| kimi-k3 | 0.495 | 0.499 | 0.497 | 0.497 | 0.0020 | 9 / 9 / 9 |
| glm-5.3-flash | 0.488 | 0.483 | 0.481 | 0.484 | 0.0036 | 10 / 10 / 10 |

**Adjudicator.** Using a single judge removes judge identity as a source of differences between systems, but the ordering may still depend on which judge was chosen. We therefore re-scored a fixed replay sample of 200 episodes per system with three alternative judges, keeping transcripts, scenes and rubric unchanged (Tables 24 and 25). The ordering of the ten systems is largely preserved: Spearman's $\rho$ with the reference ordering is 0.976 to 0.988, and no system moves by more than one rank under any judge. Every rank change occurs between systems that the reference judge separates by at most 0.015, namely the first and second, the second and third, and pairs among the three systems within 0.019 of one another discussed in §4.2.1; no change crosses a gap that the paper treats as meaningful. Absolute levels shift more than orderings. `qwen3.8-max` as judge scores every system higher, by 0.037 on average, while leaving the order nearly unchanged, so judge leniency acts mostly as an offset. The judges also differ in which of the two leading systems they place first: `gpt-6-astra` and `gemini-3.8-pro` put `gpt-6-astra` ahead of `claude-opus-5.5`, whose reference lead is 0.008, and we accordingly do not claim an order between them. A further question is whether judges favor systems of their own family, and the replay suggests that they do, modestly. Relative to the remaining systems, `gpt-6-astra` as judge raises its own composite by 0.023, `qwen3.8-max` raises its own by 0.013, and `gemini-3.8-pro` raises `gemini-3.8-flash` by 0.020. The reference judge belongs to the same family as three systems on the roster. If it favored them to a similar degree, those three should fall relative to the others under every alternative judge; they change instead by $+0.004$, $-0.009$ and $-0.005$, with no consistent sign. Any family advantage under the reference judge is therefore no larger than the differences this paper already declines to interpret, such as the 0.008 between the two leading systems, and it leaves the remaining order intact. An earlier control, with eight models under a previous specification, compared two judges from the same family (Figure 13). We include it for context; it does not establish invariance for the present experiment.

**Extraction model.** The latent-state and event-chain records are themselves produced by a model. With the judge held fixed, we regenerated these records with two alternative extraction models and compared both the records and the resulting scores (Table 26). The comparison separates sensitivity in the descriptive layer from sensitivity in the verdict. The two layers behave differently. Event-chain records agree with the reference extraction on 0.936 and 0.954 of items, latent-state records on only 0.806 and 0.816, which reproduces the ordering of reliability found in the human review (§4.7). Yet the downstream composites barely move: the mean absolute change is 0.004 to 0.005, Spearman's $\rho$ is 0.988 for both extractors, and the largest rank change is one place. Disagreement in the latent record appears to be absorbed before it reaches the verdict, which is the behavior the design intends. The adjudicator never reads numeric degrees and judges state progression from behavior in the dialogue (§3.5.2), so the less reliable of the two descriptive components has the least direct influence on the score.

Table 24: Composite score and rank under four judges, on a fixed replay sample of 200 episodes per system with transcripts, scenes and rubric unchanged. Claude Opus 5 is the reference judge used throughout the paper.

| | Claude Opus 5 | | gpt-6-astra | | gemini-3.8-pro | | qwen3.8-max | |
|---|---|---|---|---|---|---|---|---|
| Scored system | final | rank | final | rank | final | rank | final | rank |
| claude-opus-5.5 | 0.744 | 1 | 0.730 | 2 | 0.731 | 2 | 0.780 | 1 |
| gpt-6-astra | 0.736 | 2 | 0.745 | 1 | 0.744 | 1 | 0.756 | 3 |
| claude-fable-5.1 | 0.721 | 3 | 0.724 | 3 | 0.725 | 3 | 0.757 | 2 |
| claude-5-opus | 0.632 | 4 | 0.616 | 4 | 0.640 | 4 | 0.660 | 4 |
| gpt-5.6-sol | 0.602 | 5 | 0.593 | 5 | 0.598 | 5 | 0.630 | 5 |
| qwen3.8-max | 0.564 | 6 | 0.538 | 6 | 0.582 | 6 | 0.612 | 6 |
| gemini-3.8-flash | 0.526 | 7 | 0.515 | 7 | 0.550 | 7 | 0.571 | 7 |
| deepseek-v4-pro | 0.503 | 8 | 0.486 | 8 | 0.518 | 8 | 0.536 | 9 |
| kimi-k3 | 0.497 | 9 | 0.469 | 10 | 0.493 | 9 | 0.549 | 8 |
| glm-5.3-flash | 0.484 | 10 | 0.477 | 9 | 0.485 | 10 | 0.524 | 10 |

Table 25: Adjudicator sensitivity: agreement of each alternative judge with the reference judge on the ordering of the ten systems (Table 24).

| Judge model | Spearman $\rho$ | Kendall $\tau$ | mean $\lvert\Delta$ final$\rvert$ | max rank shift |
|---|---|---|---|---|
| gpt-6-astra | 0.976 | 0.911 | 0.014 | 1 |
| gemini-3.8-pro | 0.988 | 0.956 | 0.010 | 1 |
| qwen3.8-max | 0.976 | 0.911 | 0.037 | 1 |

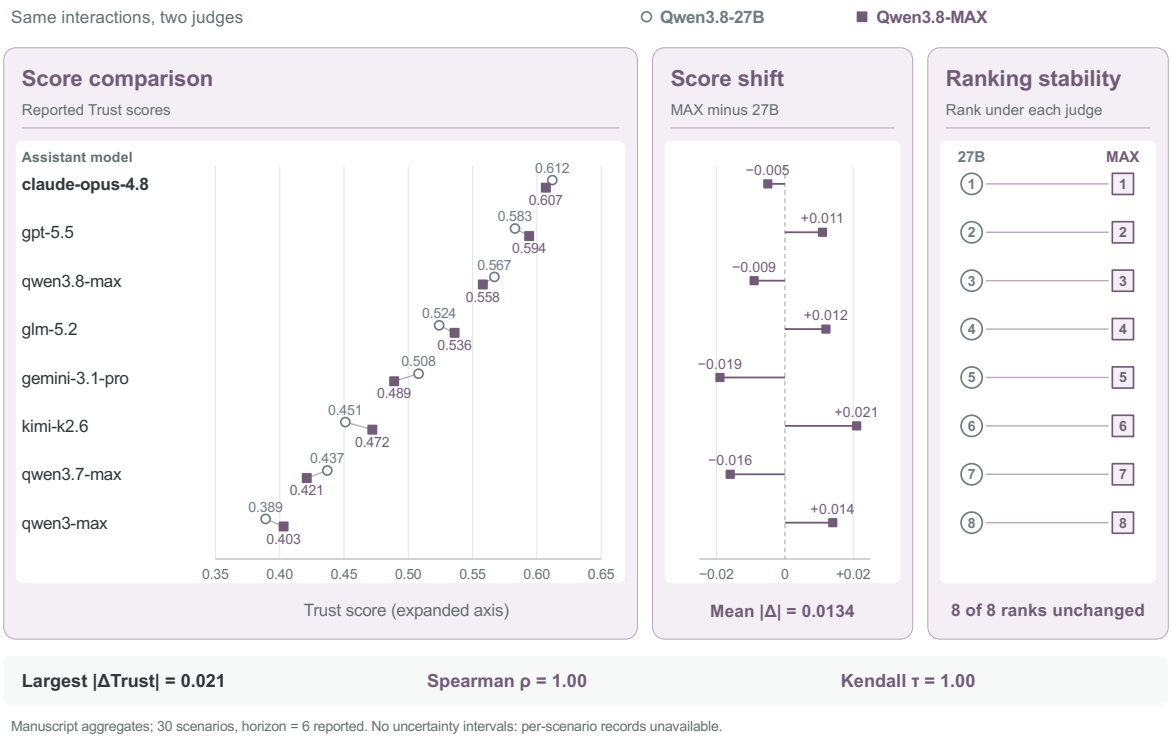



Figure 13: Judge-swap sensitivity in a separate eight-model control. The panels show reported Trust means under Qwen3.8-27B and Qwen3.8-MAX judges, signed differences (MAX minus 27B) and aggregate ranks. These descriptive means do not establish invariance for the present ten-system experiment; per-scenario records and uncertainty estimates are unavailable.

Table 26: Extraction model: agreement of the latent-state and event-chain records with the reference extraction, averaged over the ten systems, and the resulting change in downstream scores, with the judge fixed at Claude Opus 5.

| Extraction model | latent agree. | event-chain agree. | $\rho$(final) | mean $\lvert\Delta$ final$\rvert$ | max rank shift |
|---|---|---|---|---|---|
| qwen3.8-max | 0.806 | 0.936 | 0.988 | 0.004 | 1 |
| gemini-3.8-pro | 0.816 | 0.954 | 0.988 | 0.005 | 1 |

**Input format.** The surface form of a judge's input can by itself change rankings (Sclar et al., 2024). Keeping content fixed, we varied the serialization of scenes and profiles in five ways: random field order, a JSON container, a YAML container, natural-language prose and perturbed whitespace (Tables 27 and 28). Shuffling fields and perturbing whitespace change no rank at all. The three variants that change the container each swap one pair, always `deepseek-v4-pro` and `kimi-k3`, whose reference composites differ by 0.006. Format affects absolute scores more than it affects the ordering, and the largest effect comes from prose, which lowers the composite of nine systems out of ten and moves it by 0.014 on average. Presenting a scene as continuous text appears to make its constraints harder for the judge to check than presenting it as labelled fields, which is consistent with the decision in §3.2.3 to fix and version a structured serialization.

Table 27: Composite under five serializations of scenes and profiles, content held fixed, beside the reference format (structured key–value pairs in a fixed field order). Rank changes relative to the reference are marked with an asterisk.

| System | reference | shuffled fields | JSON | YAML | prose | whitespace |
|---|---|---|---|---|---|---|
| claude-opus-5.5 | 0.744 | 0.737 | 0.759 | 0.741 | 0.721 | 0.739 |
| gpt-6-astra | 0.736 | 0.737 | 0.739 | 0.732 | 0.720 | 0.739 |
| claude-fable-5.1 | 0.721 | 0.720 | 0.722 | 0.726 | 0.719 | 0.717 |
| claude-5-opus | 0.632 | 0.634 | 0.637 | 0.623 | 0.619 | 0.632 |
| gpt-5.6-sol | 0.602 | 0.594 | 0.607 | 0.603 | 0.577 | 0.600 |
| qwen3.8-max | 0.564 | 0.549 | 0.575 | 0.560 | 0.549 | 0.564 |
| gemini-3.8-flash | 0.526 | 0.530 | 0.512 | 0.528 | 0.518 | 0.522 |
| deepseek-v4-pro | 0.503 | 0.509 | 0.502* | 0.495* | 0.486* | 0.504 |
| kimi-k3 | 0.497 | 0.487 | 0.505* | 0.506* | 0.511* | 0.504 |
| glm-5.3-flash | 0.484 | 0.487 | 0.487 | 0.485 | 0.474 | 0.481 |

Table 28: Format sensitivity: agreement of each serialization with the reference format on the ordering of the ten systems (Table 27). Ties in the shuffled and whitespace variants lower $\rho$ slightly below one without any change of rank.

| Format variant | Spearman $\rho$ | mean $\lvert\Delta$ final$\rvert$ | max rank shift |
|---|---|---|---|
| Shuffled field order | 0.994 | 0.006 | 0 |
| JSON container | 0.988 | 0.007 | 1 |
| YAML container | 0.988 | 0.005 | 1 |
| Natural-language prose | 0.988 | 0.014 | 1 |
| Whitespace and indentation | 0.994 | 0.003 | 0 |

## 5 What Interactive AI Evaluation Reveals

Section 4 tested whether the predictions of Section 2 hold. This section turns to the question the instrument was built to answer: **what becomes measurable, once the interaction world is represented explicitly, that remained invisible while evaluation stopped at the response?** We organize the answer around five regularities. The first three concern initiative: where it separates contemporary systems, how its absence can pass for quality, and why it must be measured with its direction and not simply rewarded. The last two concern the interactive character of the measurement itself: endings belong to tasks, and scores belong to pairs. Each regularity is a quantity that response-level or single-score evaluation either cannot represent or represents in a way that misreads it systematically (Table 29).

Table 29: Five regularities made measurable by interactive evaluation. The second column states what an evaluation of individual responses, or of a single holistic score, would record; the third states what the instrument records instead; the fourth names the property of Section 2.3 on which the regularity chiefly depends.

| Regularity | Seen by response-level or single-score evaluation | Measured by the instrument | Property | Evidence |
|---|---|---|---|---|
| Initiative separates systems | One ordering, with ties inside 0.019 | Spans widen by stage: 0.042 for passive execution, 0.130 for responsive adaptation, 0.431 for active adaptation; profiles diverge at equal composites | Active | §4.2 |
| Passivity passes for quality | Every turn acceptable | Declining projected advancement while passive and responsive levels stay intact | Active | §4.8 |
| Initiative has a direction | Confident, specific answers read as competence | Half-given briefs are hardest; a false premise costs 0.04–0.11; fabrication peaks there and rises with competence, clarification rises with the gap | Active | §4.3, §4.4 |
| Endings reflect the task | Early stopping as failure, closure as success | Closure several to twenty times more frequent in functional settings; retention tracks $\Delta T$ | Interactive | §4.2.1, §4.4 |
| Scores belong to pairs | A fixed partner, unreported | 10.9% of variance attributable to pairing; self-play excess growing with self-similarity | Interactive | §4.5.1 |

### 5.1 What separates systems is initiative

The Agency Principle predicted that, if initiative is a capability in its own right, contemporary systems would differ most at the active end of the progression, and the results bear this out. Across ten frontier systems, passive execution spans 0.042, responsive adaptation 0.130 and active adaptation 0.431. Holding a goal and taking in information are close to saturation, updating state and adapting expression vary moderately, and the largest differences concern whether a system carries an interaction forward on its own initiative and keeps it coherent across the turns it has shaped. A single response reveals least about this part of the progression, since initiative is defined by what a turn does to the interaction and not by how well it fits the turn before.

The active stage is not a rescaled copy of the earlier ones. `deepseek-v4-pro` has the strongest passive execution on the roster and the weakest active adaptation, and `gpt-5.6-sol` combines weaker passive execution with active adaptation twice as strong; strength in passive execution does not predict initiative. Three systems separated by 0.019 on the composite differ by 0.173 in their purpose disposition and by 0.132 in how much of their initiative survives across turns. This is the information loss anticipated in Section 2.1.3,

$$q(\tau_A) \approx q(\tau_B) \qquad \text{while} \qquad J(\tau_A) \neq J(\tau_B),$$

and here it falls on the part of the progression where the systems differ. The two extensions of the protocol locate change at the same place. Cognitive architectures improve every base system most at the active stage, and adding participants degrades every system most there (§4.6); in both interventions the change is concentrated at the active end. **What distinguishes contemporary interactive systems is initiative, and a leaderboard position is a projection that hides it.**

### 5.2 Passivity can pass for quality

The Movie Evaluation Theory predicts a class of failure that no evaluator of individual responses can register, and the instrument observes it repeatedly: episodes in which every turn satisfies goal maintenance, information uptake, state progression and expression alignment, while projected advancement declines and structural repetition rises. Read through the Agency Principle, these episodes have a simple description. They are what a passive executor looks like when it is evaluated turn by turn. Every reply is appropriate to what preceded it, and none changes where the interaction is going. An evaluator computing $q_t = f(O_t, O_{<t})$ scores each of these turns as acceptable, and is right to do so; the failure belongs to the sequence. **A model without initiative is an executor, however fluent, and a turn-level score cannot tell an executor from an interlocutor.**

The same failure is easy for human reviewers to miss. Auditing the record turn by turn, they were more lenient than the rubric where a reply stayed warm and stylistically varied while repeating the same move, and they credited a character that restated one demand in new words for eleven turns with

long-horizon coherence. The most economical reading is that momentary pleasantness was taken for progress. Whatever its source, the pattern has a corollary for automated and human evaluation alike: **the pleasantness of an exchange and its structural progress are distinct quantities and must be scored separately**. It also raises a question about training that the present data cannot settle. Per-turn preference optimization rewards replies that are safe and agreeable whether or not they move the interaction, and it is one plausible way in which passivity could be learned.

### 5.3 Initiative cuts both ways

In the terms of the Data Iceberg Theory, the specification a system receives belongs to the observable surface, and what the specification omits or misstates belongs to the interaction world beneath it. Varying the specification systematically reveals a structure that a design with only complete and empty briefs would have misplaced. Difficulty is not monotone in the amount of information withheld. All three systems examined perform worst when the specification is half given, and the intermediate level carries about six times the within-scenario variance of the other two. Splitting that level locates the cause: a concealed gap costs almost nothing, while a false premise that conflicts with the rest of the specification costs every system between 0.04 and 0.11. For contemporary systems, an incorrect brief is a considerably harder problem than an incomplete one.

How systems meet a gap shows that action taken on a system's own account has a direction as well as a magnitude. Asking for what is missing and inventing it are both such actions, and they lead to different places. Clarification rises steadily with the amount of information withheld; it is constructive initiative, a change in the interaction that allows the gap to be closed. Fabrication peaks at the half-given level, where scores are lowest, and on this roster it rises with competence in passive execution, the three systems strongest at that stage fabricating most. It moves the interaction in a direction the evidence does not warrant. The roster also shows that fabrication is not the initiative the rubric credits: `deepseek-v4-pro` fabricates at the third-highest rate while having the weakest active adaptation of all ten systems. One reading is that a system which grasps a situation well enough to infer across a gap tends to do so instead of asking, and that this confidence comes from passive execution, not from the capacity to move an interaction forward. **Self-directed action cuts both ways, and only its warranted form counts as initiative.** This is why the active levels of the rubric are conditioned on evidence and on the earlier stages. An evaluation that rewards confident, specific answers without representing what the specification actually supplied would record unwarranted invention as competence. Three consequences follow. The correctness of a brief should be varied separately from its completeness; filling, asking and inventing should be recorded alongside the capability scores they would otherwise contaminate; and any attempt to cultivate initiative through training has to constrain its direction as well as its amount.

### 5.4 Endings carry the shape of the task

When episodes end because the interaction succeeds or stops being worth continuing, the way they end becomes a measurement, and what it measures is partly the task. Successful closure is the minority outcome for every system, so that an episode's ending carries no verdict by itself. Closure is also several times more frequent in functional than in non-functional settings, more than twenty times for `gemini-3.8-flash`. A companionship scene poses no objective to be closed, and a closure rate pooled across settings penalizes a relational disposition for not completing objectives it was never given. **Outcome measures such as completion, closure and survival reflect the shape of the task at least as much as the competence of the participant**, and they can be interpreted only within a level of interaction purpose.

Robustness, too, follows disposition more closely than strength. In the hardest cell of the cube, where the world departs from real rules, information is insufficient and there is no task to serve, `gpt-5.6-sol` retains 0.841 of its episodes and `claude-5-opus`, stronger on the composite, retains 0.659. Retention tracks $\Delta T$, the gap between a system's functional and non-functional performance. When there is neither a task to defend nor enough information to absorb, the strengths measured at the passive and responsive stages have little to act on, and the systems that hold up are those whose competence does not depend on an instrumental frame. The criterion for success makes the link to initiative explicit. Closure requires that the dominant path be an advance or maintenance, so a successful ending is an interaction carried forward, or deliberately held, and then brought to a close; in relational settings this form of active adaptation shows itself as sustaining an exchange, and a closure rate cannot see it.

### 5.5 Every score belongs to a pair

The most general regularity concerns the unit to which a score is attributed. In a benchmark with a fixed simulated user the partner is invisible, yet its behavior enters every number the benchmark reports.

Scoring both participants makes that contribution measurable. Of the total variation in the $10 \times 10$ matrix, 10.9% belongs to the pairing itself, beyond what the partner or the scored system contributes alone, and above the one-tenth fixed in advance as the threshold for treating pairing as a factor. **A score obtained against a single partner is a property of the pair, reported as a property of one participant.**

This bears first on how a partner should be chosen. A partner that lowers every score, as `gemini-3.8-flash` does, separates the scored systems less than one that probes hard, and it is the spread a partner produces, correlated with its probing intensity at $+0.48$, that makes it informative. It bears also on what a simulated user is taken to be. Cooperativeness is often treated as a parameter to be set when a simulator is designed; here it is a measured property of each system in the user role, ranging only from 0.61 to 0.90, and no system behaves as an adversarial user. Self-evaluation is affected as well. A system paired with itself gains more the more closely its two roles resemble each other (correlation $-0.65$ between self-similarity and the net gap), and its strength has almost no bearing on the gain, so protocols that use a model as its own interlocutor favor systems whose partner and character styles coincide. The simulated user is part of the measuring instrument, whether or not it is reported as such. It also bears on initiative directly: partners that probe hard press the scored system at the active stage, so whether the active end of a system's profile is exercised at all depends on who sits across from it. The dependence is not symmetric. In the user role, a system's score is almost entirely its own, with 94.4% of the variance attributable to the system playing the user and 2.0% to the pairing; on the assistant side the pairing component is five times as large. The joint component also grows with the number of participants, from 10.9% in dyads to 16.2% in four-person groups, so the attribution problem grows as evaluation moves from dyads toward multi-party settings.

**A common structure.** In each of the five cases, what a response-level or single-score evaluation omits is a systematic effect with a direction, and it does not average away as noise would. Profiles collapse into orderings, degenerating trajectories pass as acceptable, confident invention reads as competence, relational dispositions read as failures to close, and properties of pairs are credited to individuals. Reducing interaction to responses therefore biases the conclusions of an evaluation in predictable directions, and three of the five biases fall on initiative. **The dividing line in interactive AI is initiative**: single scores flatten it, response-level evaluation cannot see it, and an instrument that represents the interaction world can measure both its presence and its direction. The instrument can register these effects because it keeps apart what a single score conflates. A system may be locally fluent, psychologically implausible, causally stagnant or inconsistent over the long run, and each of these failures appears at a different interface of the instrument, in the rubric, the latent state, the event chain or the cross-turn level, where it can be located and reported without being absorbed into the others.

## 6 Toward a Science of Interactive AI Evaluation

### 6.1 From benchmark to measurement instrument

A benchmark ranks systems. A measurement instrument makes claims about what its scores mean, and those claims can fail. The difference matters for interactive evaluation because the object is complex enough that a plausible ranking can come from an instrument that measures something other than what it names. We have tried to build an instrument in the second sense, and several of its most useful results are those in which it was corrected by its own data. A summary index of the two active levels, which we had intended to report, added 0.0002 in fit once L5 was known and was dropped. The information facet was given three levels so that curvature could be detected, and it was; a two-level design would have reached the opposite conclusion with some confidence. Splitting the intermediate level then showed that the facet had merged incompleteness with incorrectness, which a revised design should separate. Cooperativeness, first placed on a scenario facet, proved to belong to the partner and was replaced. One class of disagreement with human reviewers was decided in the reviewers' favor and changed the rubric. An evaluation framework has to be prepared to publish results of this kind about itself if its other results are to carry weight. We regard the capacity to be revised by its own evidence as the defining property of a measurement instrument, and as a property that must be designed in from the start.

### 6.2 From responsiveness to initiative

Most of the history of language-model evaluation is a history of measuring responses. Reference answers, preference judgments, holistic ratings and per-turn rubrics differ in many respects, and all of them ask how good a response was given what preceded it. That question remains necessary, and the passive and responsive stages of the capability progression answer it. Interactive AI raises a further question, whether a model changes the interaction world or only responds within it, and the results of this paper suggest that it is now the more informative of the two. Passive execution is close to saturation across the

frontier systems we tested, and most of the variation among them lies at the active end (§5.1).

Measuring initiative matters for a reason that goes beyond ranking. Initiative is the precondition of interactive creativity: an interaction acquires new directions, conflicts and meanings only where a participant changes its course, and a model that only responds can at best make an interaction smoother. Without a measurement of initiative, an excellent executor cannot be distinguished from an interlocutor in the full sense, and the latter cannot be deliberately cultivated. The architectures of §4.6 show that cultivation is possible: both raise active adaptation more than any other stage, for every base system, and they raise it in opposite directions along the purpose facet, which only a measurement of initiative and its direction could reveal. The results also show that initiative has to be measured together with its direction. At an information gap, the systems strongest in passive execution are also the ones most likely to invent what is missing instead of asking for it (§5.3), so an evaluation that simply rewarded proactivity would reward misdirected proactivity as well.

The three properties of Section 2.3 are what make such a measurement usable. Holistic coverage of the interaction world and of a declared space of situations lets a finding about initiative state the region of situations to which it applies, and this is the basis of generalization. Measuring initiative as a quantity of its own allows it to be studied, compared and eventually optimized, which is the basis of creation. Scoring both participants over whole trajectories, with evidence attached to every judgment, makes each claim about initiative traceable and refutable, and so supplies the empirical grounding the other two depend on. **Measuring initiative is the first step from evaluating responses to evaluating interaction.**

### 6.3 Principles for interactive evaluation

The design and the results together suggest seven principles that we expect to hold for interactive evaluation generally, beyond the particular instrument described here.

- **Declare the space of situations.** Coverage should be a verifiable property of a corpus, obtained by crossing declared facets. A collection of prompts cannot say where its conclusions stop, and a design too coarse to express curvature can misplace an effect such as that of the information facet.
- **Describe before judging.** The reconstruction of what happened should be kept apart from the standard by which it is judged, so that the standard can be revised, contested or replaced while the evidence is retained.
- **Ground every inference, and prevent circular ones.** Latent variables become measurement variables when they are tied to evidence and their reliability can be assessed. An evaluator's intermediate estimates, and the labels used to sample the data, should never serve as evidence for the conclusions they help produce.
- **Measure initiative, and its direction.** Capability should be assessed along a progression that separates responding within the interaction world from acting on it. Initiative should be credited only when it rests on the earlier stages and on evidence, and constructive and misdirected initiative, such as asking and inventing at an information gap, should be recorded separately.
- **Measure the partner.** An evaluation that involves an interlocutor measures a pair. The interlocutor should be scored, reported and chosen for the spread it produces, and the effect of changing it should be estimated.
- **Report profiles with aggregates, and read outcomes within the task.** A composite is a legitimate summary when the profile it summarizes is reported beside it, and outcome measures such as closure and completion are interpretable relative to what the task asks.
- **Treat the evaluator as an object of measurement.** Agreement with human reviewers, sensitivity to the choice of judge, extractor and input format, and stability across runs are properties of the instrument, to be measured and reported with the same care as the properties of the systems it evaluates.

### 6.4 Scope of the present evidence

The evidence behind these conclusions is uneven, and the unevenness has a pattern. Causal event structure and red-line detection are reliable under audit; latent-state calibration and the two levels of active adaptation are less so, and the substitution of extraction models shows the same ordering, with latent records agreeing on about four items in five and event records on nineteen in twenty. The human study that shows this was conducted under an earlier specification, in which L1 was worded as persona maintenance and the degree ladder had more anchors, and on an earlier roster, so it fixes the shape of the reliability gradient and not its values under the current wording. The interaction matrix covers all ten

systems in both roles, which allows the pairing effect and the self-play excess to be estimated; the size of the self-play excess remains an upper bound, since systems of the same family are stylistically closer than others. The curvature of the information facet rests on three systems and the roster-level conclusions on ten. These are samples of current frontier behavior and license no claim about systems outside them.

The four controls on the pipeline, repeated passes and the substitution of judge, extraction model and input format, leave the ordering of systems essentially unchanged: the three passes agree perfectly, and under the substitutions every rank change is confined to systems the reference configuration separates by at most 0.015 (§4.9). These controls bound the dependence of the ordering on the configuration that produced it. They do not bound absolute levels, which shift by up to 0.037 with the judge and by 0.014 on average with prose serialization, so composites from different configurations should not be compared directly. Judges also favor systems of their own family by 0.013 to 0.023; the evidence that the reference judge does so is weak, but differences between Claude-family systems and the rest smaller than about 0.01 should be read with that possibility in mind. The probe of emergent behavior returned no candidate in 24,300 episodes, and whether this reflects the systems or the sensitivity of the indicator remains open. The extensions carry limits of their own. The two cognitive architectures share a family resemblance, since both reason explicitly about the partner's mental state, and the near-perfect relation between a base system's level and its gain rests on ten bases; the group experiments vary size with the standard user in every other seat, and the mixed groups report the composite and the interaction residual only. Raw scene text remains visible to the judge and may reveal the cell from which a scene was drawn, so scoring neutrality has been established for the structured labels alone, and a leakage audit is required to bound the rest. Until that audit is done, the possibility that the judge partly infers a scene's cell from its text, and scores accordingly, cannot be excluded.

### 6.5 From evaluation to learning and to persistent worlds

The record the instrument produces has the form a training signal would need. It names the level at which an interaction stalls, links a projected decline to the capability whose exercise would have averted it, and distinguishes a successful ending from an exhausted one, which a per-turn reward cannot do. Because the object a leaderboard aggregates is also the object a training loop would consume, no separately engineered reward can drift away from the evaluation. Whether training on this record improves interactive competence is untested. A test will require comparison with scalar-reward and preference-learning baselines, evidence that gains at targeted levels are not paid for at others, and close monitoring for exploitation of the evaluator, a risk already documented for evaluators used as training signals (Wang et al., 2024). The regularity of §5.2 gives such a test a concrete target: if per-turn preference optimization produces locally coherent, globally degenerating trajectories, a reward that credits structural progress should reduce them. The direction of initiative needs the same care in training as in evaluation. A reward for proactivity alone would also reward invention at information gaps, which the systems strongest in passive execution already show most (§5.3), and the dependency cascade and the evidence requirements of the rubric are the natural constraints to carry over.

The second extension is to persistent worlds, in which histories accumulate across episodes and many agents share one environment. These settings stress the variables the instrument represents, namely memory, goals that change over time, non-stationary partners, relationships among several parties and the long-run consequences of actions, and they would test the same machinery under conditions closer to those in which interactive systems are deployed. Both extensions are proposals for further study and not results of the present evaluation.

### 6.6 Closing perspective

We began from the observation that interactive systems are experienced as interactions over time, while most evaluation still measures responses. The Data Iceberg Theory identified what a response leaves out, the Movie Evaluation Theory identified when that matters, and the Agency Principle identified the capability at stake: whether a model merely executes within an interaction world or acts upon it. InteractEval turns all three into an instrument that samples situations from a declared space, reconstructs the scene, the participants' states and the causal course of an interaction, judges the reconstruction along a progression from passive execution to active adaptation, decides when an interaction has succeeded or lost its value, and scores both participants on the same terms. Evaluation of this kind is a measurement problem, and a mature discipline already exists for it: coverage can be designed, agreement estimated, dimensionality tested and sensitivity to the choice of judge measured. The evidence so far shows that the instrument separates local response quality from the quality of a trajectory, that trained reviewers endorse the distinction where the construct is structural, and that it brings into view regularities in contemporary systems that a single score would misread, the most consequential being that frontier systems now differ less in how well they respond than in whether, and in which direction, they take initiative. **To evaluate**

**interactive AI is to measure an interaction world and to ask whether a model executes within it or acts upon it; the discipline of measurement applies to both questions in full.** Claims about learning and persistent social worlds remain bounded by experiments not yet run, and we have tried to keep that boundary visible wherever the evidence is reported.

## References


Anthropic. Claude: Fable 5.1 and mythos 5.1, September 2026. URL https://www.anthropic.com/claude-fable-and-mythos-5-1.

Zhuang Chen, Jincenzi Wu, Jinfeng Zhou, Bosi Wen, Guanqun Bi, Gongyao Jiang, Yaru Cao, Mengting Hu, Yunghwei Lai, Zexuan Xiong, and Minlie Huang. ToMBench: Benchmarking theory of mind in large language models. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15959–15983, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.847. URL https://aclanthology.org/2024.acl-long.847/.

Yao Dou, Michel Galley, Baolin Peng, Chris Kedzie, Weixin Cai, Alan Ritter, Chris Quirk, Wei Xu, and Jianfeng Gao. SimulatorArena: Are user simulators reliable proxies for multi-turn evaluation of AI assistants? In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 35212–35290, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-332-6. doi: 10.18653/v1/2025.emnlp-main.1786. URL https://aclanthology.org/2025.emnlp-main.1786/.

Google DeepMind. Gemini 3.8 flash, September 2026. URL https://deepmind.google/models/gemini/flash/.

Ruth Guttman and Charles W. Greenbaum. Facet theory: Its development and current status. *European Psychologist*, 3(1):13–36, 1998.

Kashmir Hill. A teen was suicidal. chatgpt was the friend he confided in. *The New York Times*, 26, 2025.

Klaus Krippendorff. *Content Analysis: An Introduction to Its Methodology*. Sage, Thousand Oaks, CA, 2nd edition, 2004.

Robert J. Mislevy, Linda S. Steinberg, and Russell G. Almond. On the structure of educational assessment. *Measurement: Interdisciplinary Research and Perspectives*, 1(1):3–62, 2003.

OpenAI. Gpt-6-astra, September 2026. URL https://openai.com/index/gpt-6-astra/.

Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, Cambridge, 2nd edition, 2009.

Qwen. Qwen3.8-max: A new bar for coding and cowork, August 2026. URL https://qwen.ai/blog.

Georg Rasch. *Probabilistic Models for Some Intelligence and Attainment Tests*. Danmarks Paedagogiske Institut, Copenhagen, 1960.

Sahand Sabour, June M Liu, Siyang Liu, Chris Z Yao, Shiyao Cui, Wen Zhang, Xuanming Zhang, Yaru Cao, Advait Bhat, Jian Guan, et al. Human preferences are susceptible to covertly misaligned ai advice. *Proceedings of the National Academy of Sciences*, 123(38):e2600684123, 2026.

Marks Sam, Lindsey Jack, and Olah Christopher. The persona selection model: Why ai assistants might behave like humans, February 2026. URL https://alignment.anthropic.com/2026/psm/.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design, 2024.

Murray Shanahan, Kyle McDonell, and Laria Reynolds. Role play with large language models. *Nature*, 623(7987):493–498, 2023. doi: 10.1038/s41586-023-06647-8. URL https://doi.org/10.1038/s41586-023-06647-8.

Quan Tu, Shilong Fan, Zihang Tian, Tianhao Shen, Shuo Shang, Xin Gao, and Rui Yan. CharacterEval: A Chinese benchmark for role-playing conversational agent evaluation. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11836–11850, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.638. URL https://aclanthology.org/2024.acl-long.638/.

Ruiyi Wang, Haofei Yu, Wenxin Zhang, Zhengyang Qi, Maarten Sap, Yonatan Bisk, Graham Neubig, and Hao Zhu. SOTOPIA-$\pi$: Interactive learning of socially intelligent language agents. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12912–12940, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.698. URL https://aclanthology.org/2024.acl-long.698/.

Yang Xu, Xuanming Zhang, Min-Hsuan Yeh, Jwala Dhamala, Ousmane Dia, Rahul Gupta, and Sharon Li. Lh-deception: Simulating and understanding llm deceptive behaviors in long-horizon interactions. In *International Conference on Learning Representations*, volume 2026, pages 28869–28902, 2026.

Xuanming Zhang, Yuxuan Chen, Samuel Yeh, and Sharon Li. Cognition-of-thought elicits social-aligned reasoning in large language models. *Advances in Neural Information Processing Systems*, 2026a.

Xuanming Zhang, Yuxuan Chen, Samuel Yeh, and Sharon Li. Metamind: Modeling human social thoughts with metacognitive multi-agent systems. *Advances in Neural Information Processing Systems*, 38: 26133–26178, 2026b.

Xuanming Zhang, Sining Zhoubian, Yuxuan Chen, Tianyi Tang, An Yang, Sean Du, Chujie Zheng, Fei Huang, Dayiheng Liu, Gao Huang, et al. Deeper is not always better: Mitigating the alignment tax via confident layer decoding. *Empirical Methods in Natural Language Processing*, 2026c.

Yutong Zhang, Dora Zhao, Jeffrey T Hancock, Robert Kraut, and Diyi Yang. Interaction with ai companions and psychological well-being. *Nature human behaviour*, pages 1–15, 2026d.

Xuhui Zhou, Hao Zhu, Leena Mathur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, and Maarten Sap. SOTOPIA: Interactive evaluation for social intelligence in language agents. *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=mM7VurbA4r.